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SURVEY

A Survey of Indoor Localization Systems for Multi-Floor Environments

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ABSTRACT Most indoor localization systems determine only the 2D location of a user on a single floor, often overlooking multi-floor environments. This limitation reduces their effectiveness in scenarios such as emergency response in multi-floor buildings. As a result, multi-floor localization systems capable of identifying the user's floor as well have recently gained significant attention. However, designing and deploying accurate multi-floor localization systems presents unique challenges due to the increased scale and complexity of the environment. These challenges include ensuring sensor reliability, addressing signal space ambiguity caused by complex environmental structures and infrastructure configurations, and managing deployment overhead and resource requirements. Despite growing interest in this area, a comprehensive, up-to-date survey on the challenges and recent advances in multi-floor localization remains lacking. In this paper, we present the first exhaustive survey of indoor localization systems designed for multi-floor environments. We begin by highlighting the importance of floor identification through various applications and provide background on extending localization techniques from 2D to multi-floor scenarios. We then categorize and analyze the key challenges unique to multi-floor localization, including both input-level and system-level challenges. Next, we review existing solutions aimed at mitigating these challenges, with a focus on recent advancements. Finally, we discuss open problems and outline future research directions. Although significant progress has been made in indoor localization, multi-floor localization remains an active and evolving research area with numerous opportunities to bridge existing gaps and drive further innovation.

INDEX TERMS Multi-floor localization, floor identification, indoor localization, multi-floor environments, location-based services.

I. INTRODUCTION

With the increasing ubiquity of sensor-rich user devices, such as smartphones, and the widespread availability of wireless communication infrastructures, indoor localization has become integral to numerous applications [1]. It has been used in emergency response services [2], [3], evacuation management [4], social networking [5], indoor navigation [6], and location-based marketing [7], among others. In fact, the current market for indoor location-based services is valued at approximately **$58 billion** [8], highlighting its growing importance. As a result, indoor localization has been extensively studied, leading to progressively more advanced solutions aimed at enhancing the user experience. However, most existing systems focus only on **2D localization**, determining the horizontal coordinates of a user while disregarding the vertical dimension [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32]. These systems are inherently limited to single-floor environments, since they lack the capability to detect changes in the floor. Given the increasing demand for indoor location-based services in **multi-floor environments**, there is a pressing need for localization solutions that can accurately determine a user's floor, enabling seamless adoption in multi-floor buildings.

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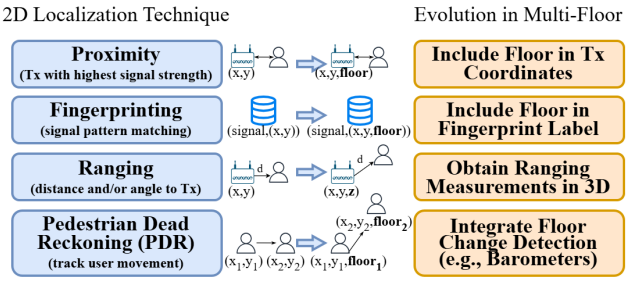


FIGURE 1. Conceptual transition from 2D to multi-floor localization by integrating vertical positioning, 3D modeling, and floor change detection.

In many scenarios, accurately identifying a user's floor is even more critical than determining their precise 2D location. For example, in **E-911 emergency calls**, having an accurate floor estimate can significantly reduce search times for first responders [2], [3]. This has been demonstrated in the San Francisco emergency response field tests [2], where providing correct floor information reduced search times by **4 to 17 minutes**. Such reductions are highly crucial, as each **one-minute decrease** in emergency response time is estimated to save over *10,000* lives per year in the USA alone [3]. Recognizing this urgency, the federal communications commission (FCC) has mandated a strict vertical positioning accuracy requirement of three meters (i.e., within one floor error) for *80%* of E-911 calls [2], [3]. The US commercial mobile radio service (CMRS) providers must comply with this requirement by **April 2025** [2], [3]. In addition, the importance of accurate floor identification extends beyond emergency response services, significantly impacting user experience in location-based customer-centric services. For instance, in multi-floor navigation, users may tolerate reasonable 2D localization errors (e.g., *up to 20 meters* [6]), but floor identification errors are far less acceptable, as they lead to incorrect floor maps and misleading navigation guidance.

However, existing 2D localization solutions cannot be directly applied to real-world multi-floor environments due to the absence of vertical positioning. In fact, multi-floor localization represents a natural but complex extension of 2D localization systems, introducing an added spatial dimension and increased complexity and scale. Figure 1 illustrates this evolution, showing an overview of how typical 2D localization techniques must be adapted or extended to multi-floor localization, resulting in a new spectrum of solutions that can incorporate floor identification. This ranges from extending traditional 2D localization techniques, such as fingerprinting [17], [19], to introducing novel approaches specifically designed for floor identification, such as leveraging phone-embedded barometers or inertial sensors to detect floor changes. Nevertheless, multi-floor localization introduces several unique challenges that must be addressed to ensure high accuracy, reliability, and practical deployability. We divide these challenges into two non-overlapping categories according to whether they concern the input measurements given to the localization system or are challenges in building the system itself, summarized as follows.

- **Input-Level Challenges**: This category involves challenges in using sensor measurements for floor change detection (e.g., barometer-based systems) and in relying on signal interpretation for floor identification (e.g., WiFi fingerprinting-based systems). On one hand, sensor-centric systems face challenges due to the effect of environmental factors (e.g., temperature and humidity) and sensor noise accumulation, leading to incorrect floor change detections [33], [34]. On the other hand, signal-based systems suffer from floor ambiguity in signal space due to complex environmental structures like *atriums* and *hollow spaces*, as well as suboptimal signal transmitter configurations, resulting in similar signal distributions across adjacent floors and reducing floor distinguishability. Physical obstructions, such as walls and furniture, can also cause similar levels of signal attenuation in both *horizontal* and *vertical* directions, further diminishing floor distinguishability.
- **System-Level Challenges**: The second category focuses on the localization system building-related challenges arising from the increased scale of multi-floor environments compared to 2D localization. These challenges include calibration and portability overhead and resource requirements. Specifically, building localization systems can involve extensive calibration data collection efforts to build location inference models, an overhead that increases substantially in multi-floor scenarios since data collection must be repeated for each floor. This process becomes especially burdensome when the system requires re-calibration for each new environment. In addition, the increased complexity and volume of the calibration data requires higher computational and storage resources, limiting the deployability of these systems on resource-constrained devices.

Several solutions have been proposed to tackle the challenges associated with multi-floor localization. For instance, to improve barometer and inertial sensor reliability, some systems combine sensor data with other localization technologies to reset error accumulation. In addition, some fingerprinting-based systems have explored clustering techniques and deep learning-based feature space mapping to reduce signal space ambiguity and enhance floor distinguishability. To mitigate deployment overhead and reduce resource requirements, certain approaches utilize crowdsourced calibration data collection and leverage lightweight models, enabling deployment on resource-limited devices. Despite these advancements, current efforts in multi-floor localization remain insufficient, leaving several open problems that require further research. *Unfortunately, the literature lacks a comprehensive and up-to-date survey*

TABLE 1. Summary of indoor 2D localization survey papers.

Survey	Contribution
Wang et al. [8]	Explores pedestrian navigation and positioning methods using smartphone inertial sensors.
Zafari et al. [45]	Reviews various localization technologies and techniques (with a focus on localization in the context of IoT) and compares different systems.
Zhu et al. [46]	Surveys fingerprinting-based localization systems that employ machine learning and intelligent algorithms.
Singh et al. [38]	Reviews machine learning-based approaches that use WiFi RSS, discussing data processing techniques and machine learning algorithms.
Feng et al. [37]	Focuses on WiFi-based localization systems that use deep learning, examining and comparing different neural network architectures.
Laoudias et al. [39]	Explores enabling technologies such as cellular and WLAN localization, WSNs, mobility estimation, and SLAM.
Tiglao et al. [40]	Examines smartphone sensing modalities and their role in localization techniques such as fingerprinting and PDR.
Alam et al. [41]	Focuses on solutions for device-free passive localization, comparing the advantages and limitations of active and passive localization methods.
Gu et al. [42]	Reviews localization approaches that incorporate spatial context, such as maps and landmarks, to improve positioning accuracy.
Lashkari et al. [43]	Discusses crowd-powered localization systems, including crowdsensing-based and crowdsourcing-based approaches.
Xiao et al. [44]	Surveys indoor localization from the device perspective, analyzing both device-free and device-based methods.
Jang et al. [47]	Discusses systems that mitigate fingerprint data collection, including SLAM, interpolation-based, extrapolation-based, and crowdsourcing-based.
Guo et al. [48]	Reviews fusion-based methods, categorizing them based on source, algorithm, and weight space characteristics.

that thoroughly examines recent advances, domain-specific challenges, and promising future research directions in multi-floor localization.

In this survey, we present the first comprehensive study of indoor localization solutions designed for multi-floor environments. Our goal is to provide readers with a deep understanding of key challenges, recent advances, and future research directions in this domain. We review several state-of-the-art systems, analyzing their proposed solutions for addressing the challenges of multi-floor localization. We also provide essential background on how common 2D localization techniques can be extended to multi-floor environments. We aim for this survey to serve as a valuable resource for both new and experienced researchers in the multi-floor localization field, guiding future developments.

A. RELATED INDOOR 2D LOCALIZATION SURVEY PAPERS

Although several existing surveys examine indoor localization from different perspectives [8], [35], [36], [37], [38], [39], [40], [41], [42], [43], [44], [45], [46], they mainly focus on advances in **2D localization systems**. In this section, we review the contributions of relevant 2D indoor localization surveys, demonstrating the need for a comprehensive, up-to-date survey on multi-floor localization to cover domain-specific challenges and recent advances. Table 1 summarizes the contributions of these related surveys.

Tiglao et al. [38] examine the role of smartphone sensors in indoor localization, discussing fingerprinting-based, propagation model-based, and dead reckoning-based systems. While they identify extending indoor localization to multi-floor environments as an *open challenge*, their survey does not explore solutions or recent advancements in this area. Lashkari et al. [41] focus on indoor localization systems that leverage crowd-powered techniques to collect calibration data, categorizing the systems based on user participation, device intervention levels, and site survey dependencies. The authors suggest extending automatic floor plan generation to multi-floor buildings as a *future research direction*, but their work does not provide an in-depth review of multi-floor localization techniques. Laoudias et al. [37] provide an extensive survey on cellular and wireless local area networks (WLAN)-based localization systems, data fusion-based approaches, mobility state estimation, localization in wireless sensor networks (WSN), and indoor simultaneous localization and mapping (SLAM). They highlight the importance of vertical positioning in emergency response scenarios, but do not offer a detailed review of multi-floor localization systems.

In addition, Xiao et al. [42] classify indoor localization approaches based on user device involvement, covering both device-free and device-based localization methods. Zafari et al. [43] focus on indoor localization in the context of the internet of things (IoT). The authors discuss multiple techniques and technologies used in indoor 2D localization and compare different solutions based on metrics such as accuracy, scalability, and localization latency. Zhu et al. [44] and Singh et al. [36] explore machine learning-based indoor localization, analyzing intelligent fingerprinting methods, data processing and augmentation approaches, different machine learning models, and transfer learning techniques.

B. MAIN CONTRIBUTIONS

The absence of a comprehensive survey that thoroughly examines the **domain-specific challenges** and **recent advances** in multi-floor localization raises the need for our survey. Overall, our survey offers the following key contributions:

- **Highlighting the importance of multi-floor localization**: We discuss various applications where floor identification is crucial, beyond just 2D location estimation. In addition, we provide background on how traditional 2D localization techniques can be extended to multi-floor environments.
- **Identifying central challenges**: We present an in-depth discussion of the domain-specific challenges faced by multi-floor localization systems, categorizing them into input-level and system-level challenges.
- **Comprehensive review of existing solutions**: We review state-of-the-art solutions designed to address these challenges, with a focus on recent advancements.

TABLE 2. Notations used in the paper.

Acronym	Definition	Acronym	Definition
2D	2-Dimensional	3D	3-Dimensional
AE	Autoencoder	AoA	Angle of Arrival
AP	Access Point	BLE	Bluetooth Low Energy
CAE	Convolutional Autoencoder	CMRS	Commercial Mobile Radio Service
CNN	Convolutional Neural Network	DAE	Denoising Autoencoders
DCGAN	Deep Convolutional GAN	DDIM	Denoising Diffusion Implicit Model
DDoS	Distributed Denial-of-Service	DL	Deep Learning
ELM	Extreme Learning Machine	FCC	Federal Communications Commission
FM	Frequency-Modulated	GAN	Generative Adversarial Network
GNSS	Global Navigation Satellite Systems	GPS	Global Positioning System
IMU	Inertial Measurement Unit	IoT	Internet of Things
KNN	K-Nearest Neighbors	LDA	Linear Discriminant Analysis
LEO	Low Earth Orbit	LoRa	Long Range
LSTM	Long Short-Term Memory	ML	Machine Learning
PCA	Principal Component Analysis	PDR	Pedestrian Dead Reckoning
PoA	Phase of Arrival	RFID	Radio Frequency Identification
RIS	Reconfigurable Intelligent Surface	RNN	Recurrent Neural Network
RSS	Received Signal Strength	RTT	Round Trip Time
SAE	Stacked Autoencoders	SLAM	Simultaneous Localization and Mapping
TDoA	Time Difference of Arrival	ToA	Time of Arrival
UWB	Ultra-Wideband	WKNN	Weighted KNN
WLAN	Wireless Local Area Network	WSN	Wireless Sensor Network

We classify the available systems based on the leveraged technology into WiFi-based [7], [47], [48], [49], [50], [51], [52], [53], [54], [55], [56], [57], [58], [59], [60], [61], [62], [63], [64], [65], [66], [67], [68], [69], [70], [71], [72], [73], [74], [75], [76], [77], [78], [79], [80], [81], [82], [83], [84], [85], [86], [87], [88], [89], [90], [91], [92], [93], [94], [95], [96], [97], [98], [99], [100], [101], [102], [103], [104], [105], [106], [107], [108], [109], [110], Bluetooth-based [6], [111], [112], [113], cellular-based [114], [115], [116], [117], [118], [119], [120], [121], [122], [123], [124], [125], [126], [127], barometer-based [128], [129], [130], [131], [132], [133], [134], [135], [136], inertial sensor-based [137], [138], [139], [140], [141], [142], hybrid approaches [143], [144], [145], [146], [147], [148], [149], [150], [151], [152], [153], [154], [155], [156], [157], [158], [159], [160], [161], [162], [163], [164], [165], [166], [167], [167], [168], [169], [170], [171], [172], [173], [174], [175], [176], [177], [178], [179], and others [180], [181], [182], [183], [184], [185], [186], [187], [188], [189], [190], [191], [192], [193], [194], [195], [196], [197], [198], [199], [200], [201], [202], [203], [204], [205], [206], [207], [208].

- **Outlining open problems and future research directions**: We outline key open problems and suggest promising future research directions to guide the development of more robust, scalable, and accurate multi-floor localization systems.

C. PAPER NOTATIONS AND STRUCTURE

Table 2 summarizes the notations used in this survey. The remainder of the paper is organized as follows.

- **Section** II discusses different applications of multi-floor localization, categorizing them into safety-centric, user experience-centric, and health-centric applications.
- **Section** III provides background on various localization techniques and explains how they are adapted or extended to multi-floor environments. These techniques include proximity-based, fingerprinting-based, ranging-based (lateration and angulation), and pedestrian dead reckoning (PDR) approaches.
- **Section** IV presents the domain-specific challenges in multi-floor localization, classifying them into input-level and system-level challenges.
- **Section** V reviews existing multi-floor localization systems, analyzing their proposed solutions to address the identified challenges.
- **Section** VI explores open research problems and future directions, outlining potential advancements to improve localization in multi-floor environments.
- **Section** VII provides concluding remarks.

II. MOTIVATION AND APPLICATIONS

In this section, we discuss applications that can benefit from incorporating reliable *floor identification* into indoor localization systems, extending them to work in multi-floor environments. We divide these applications according to their functional and societal impact into safety-centric, user experience-centric, and health-centric applications.

A. SAFETY-CENTRIC APPLICATIONS

1) EMERGENCY RESPONSE

Indoor localization systems play a vital role in allowing emergency responders to locate people in danger faster, reducing response times, and ultimately saving more lives.

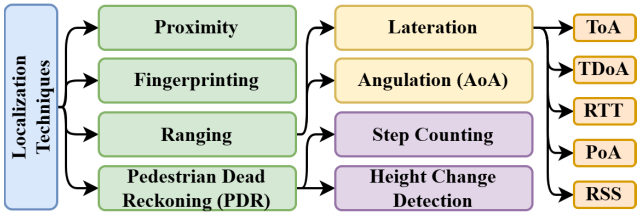


FIGURE 2. Summary of localization techniques.

Specifically, decreasing emergency response time by just **one minute** saves more than *10,000 lives annually* in the USA alone [3]. In multi-floor environments, accurately identifying the floor significantly narrows down the search area [37]. Thus, emergency responders prioritize floor-level accuracy over precise 2D location data, as knowing the correct floor makes it easier for them to locate victims [2]. As a result, the FCC has set a vertical positioning accuracy requirement of **3 meters** (i.e., floor-level accuracy) for *80%* of indoor E-911 calls in the USA, which must be met by **April 2025** [2], [3].

2) EVACUATION MANAGEMENT

Accurate floor identification can significantly help evacuate multi-floor buildings during disasters by allowing quick identification of efficient evacuation routes. This is particularly valuable when passages are blocked or when individuals lack familiarity with the building's layout [4].

B. USER EXPERIENCE-CENTRIC APPLICATIONS

1) NAVIGATION

Indoor localization systems help users navigate efficiently by identifying the shortest routes to their destination, significantly reducing travel times, especially in large-scale environments. In multi-floor settings, accurate floor identification is essential for selecting the correct floor map for navigation [6]. Thus, users have much lower tolerance for floor identification errors compared to 2D localization inaccuracies, as the former can cause the navigation system to rely on an incorrect floor map, severely disrupting the user experience [6].

2) LOCATION-BASED MARKETING AND ANALYTICS

Context-aware applications that analyze customer interests and behavior can help deliver personalized promotions and target the right audience. For example, in a shopping mall, the prolonged presence of a customer near a store often indicates an interest in the products on display. Indoor localization systems can thus help infer customer preferences based on their proximity to stores. Tailored advertisements and offers, such as discount coupons or location-based notifications, can then be delivered to the customer, enhancing user experience and increasing sales. In multi-floor malls, accurately identifying the customer's floor is crucial for pinpointing nearby stores.

3) ASSET TRACKING

Floor identification is also essential for tracking belongings and assets within indoor multi-floor environments, where common GPS tracking devices are often ineffective [25], [209]. For example, in a multi-floor parking garage, a location-based application can record both the floor and the 2D coordinates of a car in a certain parking spot. It can then track the user's walking trajectory, guiding them to easily locate their car [209].

C. HEALTH-CENTRIC APPLICATIONS

1) PATIENT AND STAFF TRACKING IN HOSPITALS

Integrating floor identification into indoor localization services can significantly enhance patient safety in large multi-floor hospitals. For example, when a patient requires immediate medical attention, hospitals often notify all staff members or assign a specific member regardless of their proximity to the patient. A more efficient solution is to rely on an indoor localization system that tracks both the floor and the 2D location of staff members. In emergencies, the system can identify and alert the nearest available staff member, taking into account floor differences, to ensure a faster response.

2) PANDEMIC MANAGEMENT AND CONTACT TRACING

During pandemic outbreaks, a multi-floor localization system can play a crucial role in public health management by identifying crowded areas on different floors and notifying users to avoid them [210], [211]. Furthermore, such systems can support contact tracing by identifying individuals who may have come in proximity with an infected person, particularly if they were on the same floor [212], [213].

III. BACKGROUND ON MULTI-FLOOR LOCALIZATION

Various indoor localization techniques have been proposed using technologies ranging from wireless communication infrastructures (e.g., WiFi) to embedded sensors (e.g., accelerometers and barometers). These techniques can be broadly divided into *proximity-based*, *fingerprinting-based*, *ranging-based*, and *pedestrian dead reckoning (PDR)*.

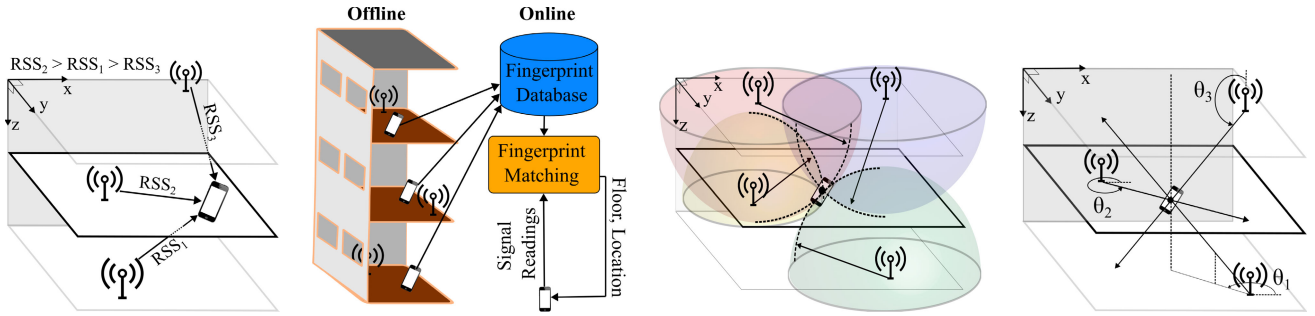
Although existing indoor localization surveys [8], [37], [38], [43] offer a comprehensive background of using these techniques for 2D positioning, they lack details on how these methods can be adapted for multi-floor environments. In this section, we highlight how multi-floor localization systems extend these techniques to accommodate the vertical dimension. Figure 2 outlines these techniques, and Table 3 summarizes their respective advantages and limitations.

A. PROXIMITY-BASED LOCALIZATION

Proximity-based approaches estimate a user's location by leveraging the position of nearby signal transmitters, where a stronger signal strength indicates closer proximity to the transmitter. Specifically, if transmitter locations are known, the location of the user is estimated to be the same as that of the transmitter with the highest received signal strength (RSS)

TABLE 3. Advantages and limitations of the different localization techniques.

Localization Technique	Advantages	Limitations
Proximity	- Calibration-free - Lightweight	- Depends on density, coverage, and transmitter availability - Highly affected by signal interference and noise
Fingerprinting	- Works with off-the-shelf devices - High accuracy even in non-line-of-sight	- Fingerprinting data collection overhead - Affected by changes in environment and infrastructure
Lateration: ToA	- No data collection overhead - High accuracy in line-of-sight	- Requires synchronizing transmitters and receivers - Severely affected by multipath and non-line-of-sight
Lateration: TDoA	- No data collection overhead - High accuracy in line-of-sight - Does not require synchronizing transmitters and receivers	- Requires synchronizing transmitters with each other - Severely affected by multipath and non-line-of-sight
Lateration: RTT	- No data collection overhead - High accuracy in line-of-sight - Less synchronization efforts than ToA and TDoA [21]	- Higher latency than ToA and TDoA [21] - Severely affected by multipath and non-line-of-sight
Lateration: PoA	- No data collection overhead - Higher accuracy than time-based lateration methods [217] - Does not require synchronization	- Effect of phase measurement ambiguity [217] - Severely affected by multipath and non-line-of-sight
Lateration: RSS	- No data collection overhead - Does not require synchronization	- Less accurate than other lateration methods [218], [219] - Less accurate than RSS fingerprinting [33], [220] - Relies on empirical environment-specific parameters
Angulation: AoA	- No data collection overhead - Does not require synchronization - Requires fewer transmitters than lateration - High accuracy at short distance from transmitters [221]	- Low accuracy at large distance from transmitters [222] - Requires line-of-sight for optimal performance - Requires complex hardware for angle measurements [223]
PDR: Step Counting	- No data collection overhead - No infrastructure deployment requirements - Suitable for continuous tracking	- Requires initial position - Error accumulation due to sensor drift and noise - Affected by device orientation and user behavior - Limited in detecting elevation changes
PDR: Height Change Detection	- No data collection overhead - Can identify stairs/elevators useful for floor change detection	- Requires initial floor - Error accumulation due to sensor drift and noise - Affected by device orientation, user behavior, and environmental conditions - Might require additional infrastructure deployment (e.g., external barometers [135], [136])



(a) Proximity-based localization. (b) Fingerprinting-based localization. (c) Lateration-based localization. (d) Angulation-based localization.

FIGURE 3. Extending the different localization techniques to multi-floor environments.

or as a weighted average of the positions of the strongest transmitters, where the weight depends on the RSS. These methods typically rely on a dense network of short-range transmitters, such as Bluetooth beacons, WiFi access points (APs), and cellular small cells.

1) EXTENDING TO MULTI-FLOOR

Proximity-based approaches can be directly adapted to multi-floor environments if the height or floor of the transmitter is included in its location information (Figure 3a) [6], [47], [50], [76], [117], [118]. Furthermore, some systems propose proximity-based heuristics for floor identification, such as selecting the floor with the highest number of visible transmitters or the one whose visible transmitters have the largest average RSS [47], [50], [76].

2) ADVANTAGES AND LIMITATIONS

Although proximity-based approaches are calibration-free and lightweight, their accuracy is heavily dependent on the density and range of transmitters [43]. Moreover, temporary unavailability, interference, or noise can distort the signal strength, often making the strongest signal correspond to a transmitter that is not the closest, affecting accuracy.

B. FINGERPRINTING-BASED LOCALIZATION

Fingerprint-based approaches rely on the fact that signal characteristics, e.g., RSS, vary with distance from the transmitter and are influenced by physical obstructions. Thus, the location of the user can be inferred using signal measurements and prior knowledge of the signal distribution at different locations in the environment. Specifically, location-tagged signal measurements can be collected from different transmitters to create a ''fingerprint database''. This database is then used to build a model that matches real-time signal measurements with collected fingerprints to estimate the user's location.

Fingerprinting consists of two main phases, an **offline phase** for building the fingerprint database and preparing the model, and an **online phase** for fingerprint matching. After the offline phase, the database can be stored to be used directly in the online phase with matching algorithms such as k-nearest neighbors (KNN) [114] or weighted KNN (WKNN) [51], or used to train a machine learning (ML) neural network for online inference [120].

1) EXTENDING TO MULTI-FLOOR

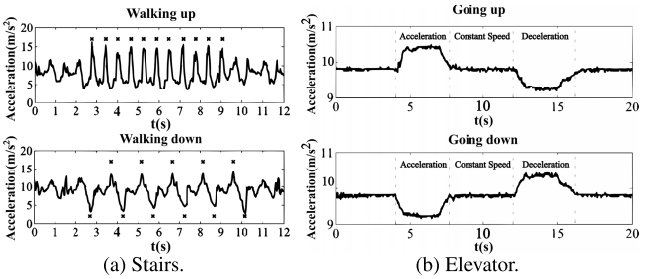
Fingerprinting can be extended to multi-floor environments by including floor labels for the signal measurements collected to build the fingerprint database (Figure 3b) [120]. If the system provides 2D location, the floors must be divided into discrete reference points, with each point surveyed to collect floor- and 2D location-tagged measurements. To determine the location, either a separate 2D location estimation model is built for each floor and the floor is first identified, followed by the 2D location [55], or a single model obtains the floor and the 2D location simultaneously [57].

2) ADVANTAGES AND LIMITATIONS

Fingerprinting is one of the most common approaches for multi-floor localization due to its effectiveness in obtaining high accuracy using off-the-shelf devices and its ability to work well in non-line-of-sight conditions [221]. However, the overhead of offline fingerprint data collection, which significantly increases in the case of multi-floor environments, is a considerable drawback. Moreover, changes in the environment and infrastructure cause the fingerprint database to become less representative of the environment over time, requiring periodic updates to the fingerprint database [27].

C. RANGING-BASED LOCALIZATION

Ranging-based approaches estimate a user's location by measuring the distance or angle between the user's device and signal transmitters with known locations. These techniques are divided into *lateration* and *angulation* depending on whether distance or angle measurements are used, respectively.



(a) Stairs. (b) Elevator.

FIGURE 4. Vertical motion detection using accelerometer readings [138].

1) LATERATION

Lateration methods calculate the distances from signal transmitters based on certain signal measurements. These measurements include the time of arrival (ToA), time difference of arrival (TDoA), round trip time (RTT), and phase of arrival (PoA) [43]. In addition, propagation models can be used with RSS measurements to estimate the distance [222].

a: EXTENDING TO MULTI-FLOOR

Lateration can be used in multi-floor environments if we know the 3D coordinates of the transmitters. In this case, the distance measurements define spherical regions centered around the transmitters, and the location can be obtained by estimating the intersection area of these spheres (Figure 3c) [77], [117], [118], [158]. As such, at least four transmitters are required to obtain the user's location.

b: ADVANTAGES AND LIMITATIONS

While lateration does not require data collection efforts, it suffers from different limitations depending on the signal measurement used to estimate the distance (Table 3). Moreover, a significant drawback of lateration-based methods is their susceptibility to non-line-of-sight propagation and multipath effects, where signals take longer paths than the actual distance between the transmitter and the receiver, reducing accuracy [223].

2) ANGULATION

Angulation determines the location of the user by leveraging the angular direction of signals relative to a reference direction. Typically, antenna arrays at the receiver are used to calculate the angles of incoming signals [220]. This approach relies on the signal's angle of arrival (AoA), where a single AoA measurement confines the user's location to a straight line originating from the transmitter.

a: EXTENDING TO MULTI-FLOOR

At least three transmitters are required to estimate the 3D coordinates of the user based on the intersection of the AoA straight lines. These coordinates can then be used to determine the location in a multi-floor environment (Figure 3d).

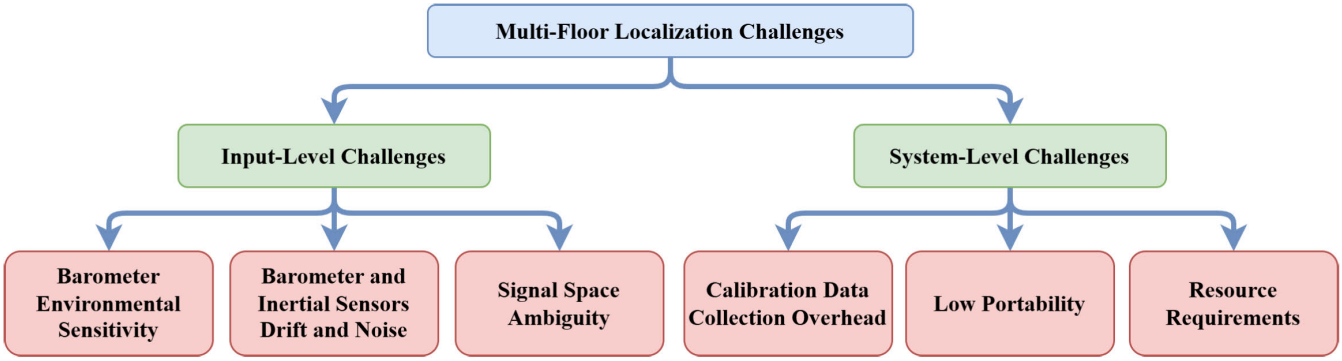
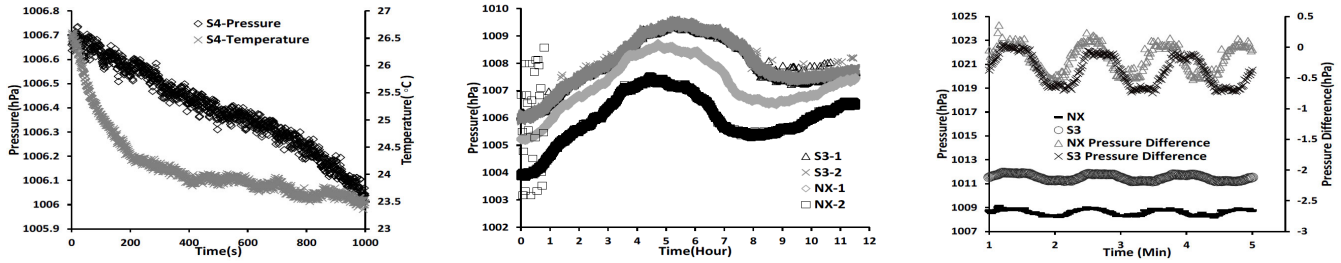


FIGURE 5. Taxonomy of the challenges facing multi-floor localization systems.



(a) Effect of temperature. (b) Change in absolute measurements with time. (c) Reliability of pressure difference.

FIGURE 6. Unreliability of absolute barometer pressure measurements compared to relative measurement differences for floor change detection [130].

b: ADVANTAGES AND LIMITATIONS

Angulation provides high accuracy when the distances between the receiver and the transmitters are short [218]. However, it requires complex hardware to measure the angular direction of the signal and necessitates line-of-sight propagation conditions for optimal performance.

D. PEDESTRIAN DEAD RECKONING

Pedestrian dead reckoning (PDR) approaches track user movement using inertial sensors such as accelerometers, gyroscopes, and magnetometers to measure acceleration, orientation, and heading direction [33]. However, inertial sensors embedded in user devices are prone to noise, leading to significant error accumulation when used directly to calculate the distance moved, where the error increases cubically over time [33], [139]. To address this, PDR employs step counting, which involves detecting a walking step, estimating step length (walking speed), and determining walking direction [224]. This substantially reduces error accumulation, making the error linear in the number of steps.

1) EXTENDING TO MULTI-FLOOR

PDR step counting can be extended to multi-floor environments by incorporating height change detection, leveraging unique patterns in inertial sensor measurements to track vertical motion (Figure 4). For example, stair usage creates distinct spikes in accelerometer readings (Figure 4a), while elevator usage produces a three-stage pattern in accelerometer readings: acceleration, constant speed, and deceleration (Figure 4b) [138]. Furthermore, barometer sensors can be integrated into PDR systems to detect changes in altitude based on air pressure, as air pressure decreases with increasing height [143], [144], [145], [146], [147], [148], [149].

2) ADVANTAGES AND LIMITATIONS

Although PDR does not require infrastructure deployment and supports continuous tracking for navigation applications, it requires an initial position estimate, which may not always be available. Moreover, sensor noise and drift cause error accumulation over time, limiting PDR's suitability for long-distance location tracking unless the absolute location is periodically re-initialized to correct the error [33]. Furthermore, sensor readings are influenced by device orientation and user activities, e.g., texting, making it challenging to isolate the effects of floor changes [182].

IV. MULTI-FLOOR LOCALIZATION CHALLENGES

In this section, we present a structured taxonomy of the core domain-specific challenges in designing multi-floor localization systems (Figure 5). We divide these challenges according to whether they are related to the system's input or are challenges in implementing and deploying the system into two non-overlapping categories: **Input-Level Challenges** and **System-Level Challenges**. In Section V, we discuss the different solutions proposed to address these challenges.

A. INPUT-LEVEL CHALLENGES

These challenges relate to issues in the quality and ambiguity of input measurements used for floor identification. They affect both **sensor-based** systems (e.g., floor change detection using barometer and inertial sensors) and **signal-based** systems (e.g., WiFi, Bluetooth, and cellular fingerprinting). Key input-level challenges include the sensitivity of barometer measurements to environmental conditions, the error accumulation caused by sensor noise and drift, and the signal space ambiguity (e.g., due to atriums or suboptimal signal transmitter placement).

1) BAROMETER ENVIRONMENTAL SENSITIVITY

The straightforward use of barometers for floor identification in multi-floor localization systems is based on the principle that air pressure decreases with altitude. Ideally, each floor should correspond to a distinct absolute pressure value, allowing systems to estimate the user's vertical position, e.g., by calculating height above sea level or building an air pressure fingerprinting database. However, absolute barometer readings are unreliable for this task due to their sensitivity to environmental conditions, especially in case of device heterogeneity [130]. Specifically, environmental conditions such as temperature, humidity, and weather changes significantly influence barometer readings even when the user's height remains constant. For instance, increasing the ambient temperature decreases air pressure which can alter the pressure readings observed at the same height. This variability can occur across different times of the day, as shown in Figure 6, which compares barometer and temperature sensor readings on a Samsung Galaxy S4 (S4) smartphone at a constant floor [130]. As seen in the figure, the absolute pressure readings significantly vary with temperature, making it difficult to rely on absolute pressure measurements for floor identification.

In addition, hardware and software heterogeneity across user devices introduces inconsistencies in absolute barometer measurements, even when taken at the same location. Figure 6b shows absolute pressure measurements collected at the same location using two Samsung Galaxy S3 (S3) and two Google Galaxy Nexus (NX) smartphones [130], illustrating that absolute pressure measurements can differ significantly, leading to a difference of up to around **10m** in height estimation [130]. Temporal variations exacerbate the problem, where, for the same device, pressure measurements change with time of day due to slow atmospheric pressure changes, making it difficult to define a stable absolute pressure for any given floor. This device and time dependence poses a substantial challenge for generalization and public dataset development, especially in crowdsourced environments.

As a result, recent systems rely instead on short-term relative pressure differences during user movement between floors. Figure 6c shows that while absolute measurements differ between devices, the relative pressure changes between two adjacent floors remains consistent, making it a more robust feature for detecting floor transitions [130].

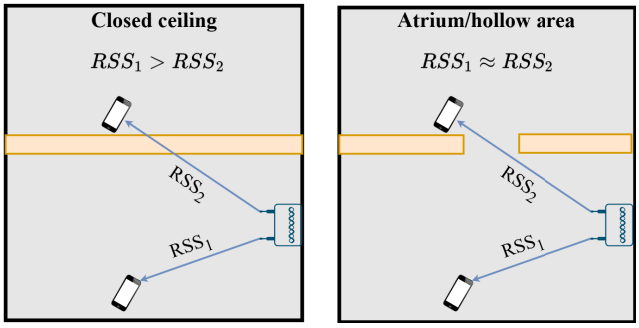


FIGURE 7. Effect of atriums and hollow spaces. Signals overlap between floors due to the absence of ceiling attenuation.

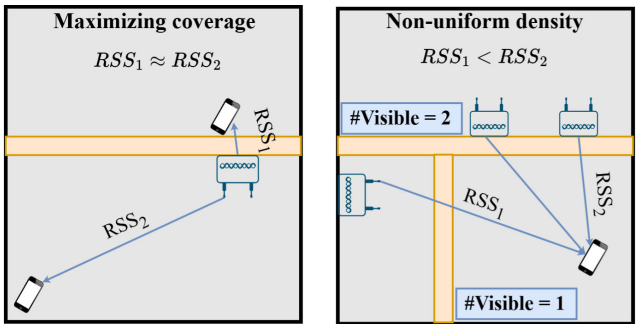


FIGURE 8. Effect of transmitters placement and density. Ceiling-mounted transmitters are typically employed to *maximize coverage*, resulting in signal overlap across floors. Furthermore, non-uniform density can cause stronger signals and more visible transmitters from unintended floors.

2) BAROMETER AND INERTIAL SENSORS DRIFT AND NOISE

Off-the-shelf devices typically include low-cost barometers and inertial sensors designed for general-purpose activity recognition rather than fine-grained localization. These sensors suffer from manufacturing imperfections and limited precision, leading to two major issues: drift, which is the gradual accumulation of small measurement errors over time, and noise, which introduces short-term fluctuations in readings.

For example, barometric drift can occur due to sensor bias and temperature-dependent behavior within the sensor hardware, which gradually shifts the pressure baseline. Barometric noise, on the other hand, arises from minor fluctuations in pressure readings caused by internal sensor instability, mechanical vibrations, or the physical handling of the device (e.g., hand movement or placement in a pocket or bag), which introduce frequent disturbances unrelated to elevation change. Similarly, inertial sensor noise results from jitter and random motion artifacts picked up during walking, stair climbing, or small hand tremors, making it difficult to distinguish between true elevation change and noise.

In multi-floor localization scenarios, these issues are particularly problematic. Drift or noise in vertical measurements can lead to false floor change detections, where the system mistakenly changes or fails to update the user's floor level. This can result in the user being tracked on

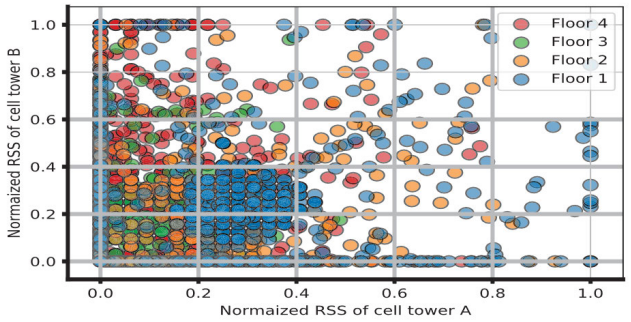


FIGURE 9. Effect of large signal coverage. Due to the wide transmission range of cell towers, signal strength values overlap across floors, reducing floor distinguishability [120].

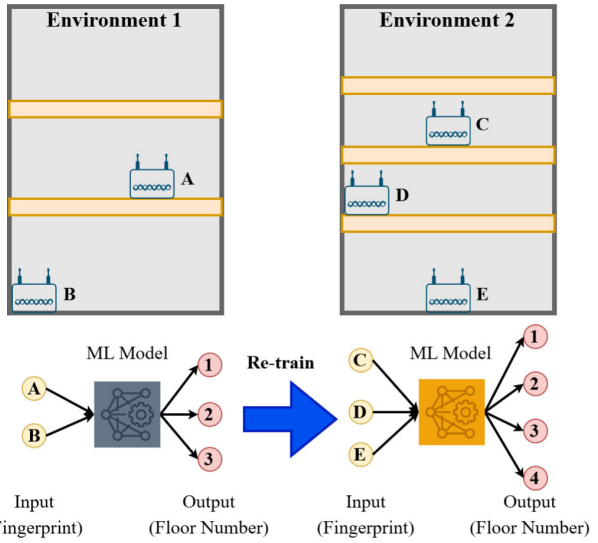


FIGURE 10. The low portability challenge. Systems built specifically for Environment 1 need to be re-trained when deployed in Environment 2 due to changes in environment structure and transmitter configuration.

the wrong floor, causing rapid accumulating positional errors. These challenges often necessitate frequent sensor recalibration or the use of external anchors, such as detected landmarks, floor maps, or fusion with more stable signals like WiFi fingerprints, to maintain localization accuracy over time [33], [34].

3) SIGNAL SPACE AMBIGUITY

Signal space ambiguity is a key challenge facing fingerprinting-based multi-floor localization systems that rely on unique signal signatures to differentiate between floors. This ambiguity arises when signal measurements from different floors become indistinguishable or overlapping, making it difficult for the system to correctly identify the user's floor.

One primary cause of signal space ambiguity is open environment structures, as seen in malls, airports, and office complexes that feature **atriums, open staircases, or hollow vertical spaces** [153]. In such environments, indoor radio signals (e.g., WiFi) can propagate freely between adjacent floors with minimal attenuation, retaining higher consistency across floors compared to signals that pass through concrete ceilings (Figure 7). This reduces spatial variability in the signal characteristics, e.g., RSS, a critical cue for distinguishing floor levels in fingerprinting-based systems [148].

Moreover, signal transmitter placement and density play a crucial role in shaping the signal landscape (Figure 8). Transmitter placement often prioritizes maximizing coverage, e.g., placing transmitters on ceilings, but this inadvertently causes similar signal profiles on the floor above, where the distance to the transmitter is still relatively small. The signal loss from the ceiling may be offset by shorter propagation distance, making it difficult to distinguish floors. Furthermore, non-uniform transmitter deployment density, where some regions have denser coverage than others, can lead to misleading proximity-based features. A user located on one floor might receive stronger signals from transmitters on a different floor (Figure 8), inducing signal space ambiguity [47], [154].

In addition, the relatively large coverage radius of certain transmitters, such as outdoor cellular towers, can cause signal space ambiguity. These towers are designed to span wide areas, resulting in very similar signal patterns across multiple floors. When floor-specific signal attenuation is small, the impact of moving vertically becomes as subtle as the effect of walls and furniture during horizontal movement within the same floor. This similarity reduces floor distinguishability in signal space, as vertical and horizontal movements have similar effects on the signal [120], [123]. Figure 9 shows RSS measurements from two cell towers taken on four different floors [120]. As seen, the RSS values have significant overlap between floors due to the large coverage of the towers, making floor differentiation challenging.

Overall, this ambiguity challenges the assumption that each floor has a unique signal fingerprint. It can lead to misclassifications, especially when signal overlap is high, and reduces the robustness of floor identification, particularly in large or open multi-floor environments.

B. SYSTEM-LEVEL CHALLENGES

These challenges refer to the issues facing the implementation and deployment of multi-floor localization systems rather than the input measurements they receive. This includes the overhead of deploying these systems at scale, portability to new environments, and storage and computational resource requirements.

1) CALIBRATION DATA COLLECTION OVERHEAD

Although fingerprinting-based systems remain among the most accurate approaches for multi-floor localization, their performance depends heavily on the availability of a dense and high-quality fingerprinting database. This database is typically manually collected through a site survey process which involves recording signal characteristics at different

reference points throughout the environment. In multi-floor environments, the calibration burden increases significantly with the number of floors, as each floor must be surveyed independently to capture floor-specific signal variations (Section III-B) [120], [124]. This labor-intensive process becomes a major scalability bottleneck, especially in large environments such as shopping malls, hospitals, and university campuses. For example, surveying a 10-floor building with 50 reference points per floor results in 500 distinct locations to be measured, often requiring days of manual effort. The cost and time associated with such calibration make fingerprinting-based systems less appealing for practical deployments.

This overhead is further increased with the adoption of deep learning (DL) models for inference in recent multi-floor localization research [49], [55], [120], [124]. While DL models can capture complex patterns in signal data and potentially improve generalization, they are typically data-hungry, requiring large training datasets to avoid overfitting and to perform robustly across diverse environments. Provisions to reduce calibration data collection overhead without compromising performance are essential for large-scale, practical, and maintainable real-world deployment of multi-floor localization systems.

2) LOW PORTABILITY

The low portability of many multi-floor localization systems represents a significant barrier to real-world deployability and scalability. This challenge arises when a system deployed in one environment cannot be directly redeployed in a new setting without significant modifications. In particular, many fingerprinting-based systems are tightly coupled with the physical and signal characteristics of the deployment environment, making redeployment costly and time-consuming. A common architectural choice in these systems is to design and train ML models tailored to a specific number of inputs and outputs. For example, a floor classifier trained on data from m transmitters in an n-floor environment learns to map an m-dimensional input vector to n discrete output classes representing the floors [49]. While effective in that particular environment, such systems fail to generalize when transferred to a new location with a different number or deployment configuration of transmitters, or different floor layouts.

Specifically, as depicted in Figure 10, redeploying the system in a different environment requires re-building the model based on the new configuration, which necessitates repeating the calibration data collection and training process from scratch. This retraining overhead not only consumes time and resources, but also limits the practicality of deploying such systems at scale, especially in dynamic environments like shopping malls, hospitals, or multi-building campuses where changes are frequent. Designing systems that can minimize reconfiguration effort and maintain accuracy without re-collecting data is challenging, where achieving high portability is critical for widespread, low-cost adoption of multi-floor localization systems in real-world scenarios.

3) RESOURCE REQUIREMENTS

Achieving high localization accuracy in multi-floor environments often comes at the expense of substantial resource requirements, including computational power, memory, and energy consumption. This challenge becomes especially significant in systems deployed on resource-constrained devices, such as smartphones or wearables, where limitations on CPU, RAM, storage, and battery life restrict the feasible complexity of algorithms. For instance, fingerprinting-based systems are known for their heavy storage and processing needs as they often rely on maintaining large fingerprint databases for use with matching algorithms like KNN or probabilistic inference. To improve accuracy, many recent systems adopt DL models, introducing additional computational burdens during both training and inference. These models can contain millions of parameters, requiring hardware acceleration (e.g., GPUs) for efficient inference. Deploying such models on mobile devices may lead to slow response times and increased power consumption, degrading user experience.

As a result, many practical systems outsource location estimation computations to a cloud server, where the user device only collects and transmits the input data to a remote server, which returns the estimated location [150]. While this approach reduces on-device resource usage, it introduces dependence on network connectivity and may result in higher localization latency. Moreover, cloud-based solutions raise privacy concerns, as user location data must be transmitted and processed externally. On the other hand, on-device localization offers privacy and latency advantages, as the full localization pipeline, from data collection to inference, is handled locally. However, achieving this requires careful resource optimization.

V. INDOOR MULTI-FLOOR LOCALIZATION SYSTEMS

In this section, we review the current indoor multi-floor localization systems, dividing them according to the technology they rely on into WiFi-based, Bluetooth-based, cellular-based, barometer-based, inertial sensor-based, hybrid, and others. For each group, we discuss how the systems address the domain-specific challenges detailed in Section IV. We then study the trade-offs across the different used technologies in multi-floor localization, comparing them based on different performance metrics, and providing insights on their suitability for different scenarios. Figure 11 summarizes the different solutions proposed to address the challenges, indicating the corresponding systems.

A. WIFI-BASED SYSTEMS

1) WIFI IN MULTI-FLOOR LOCALIZATION

WiFi APs are available in almost all indoor environments, making WiFi one of the most widely used technologies for indoor multi-floor localization. Different localization techniques leverage WiFi signals for multi-floor localization, the most common of which are proximity-based methods and RSS fingerprinting. Here, we review systems that depend

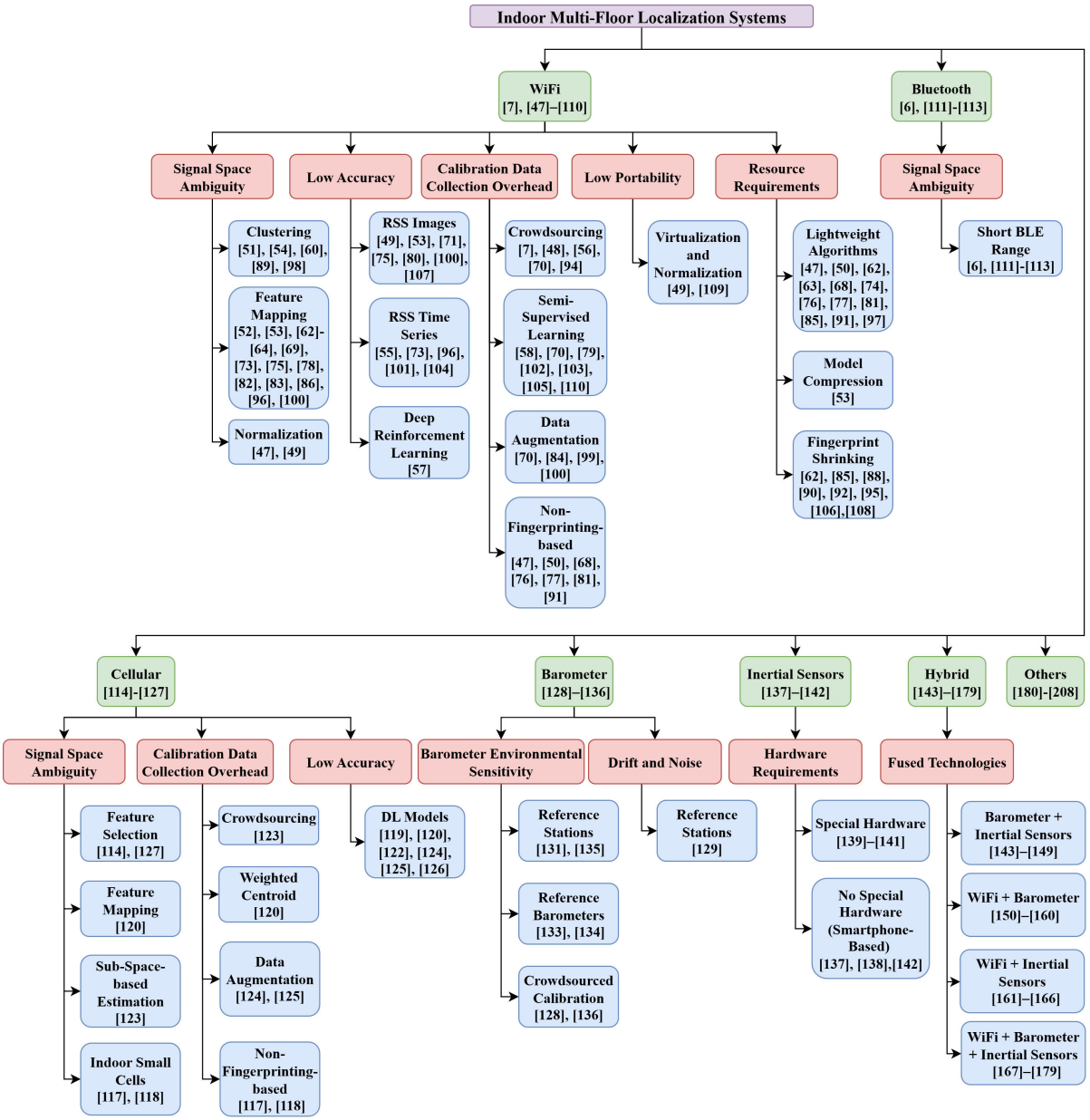


FIGURE 11. Taxonomy of the proposed solutions for the multi-floor localization challenges, categorized by the used technologies.

only on WiFi [7], [47], [48], [49], [50], [51], [52], [53], [54], [55], [56], [57], [58], [59], [60], [61], [62], [63], [64], [65], [66], [67], [68], [69], [70], [71], [72], [73], [74], [75], [76], [77], [78], [79], [80], [81], [82], [83], [84], [85], [86], [87], [88], [89], [90], [91], [92], [93], [94], [95], [96], [97], [98], [99], [100], [101], [102], [103], [104], [105], [106], [107], [108], [109], [110], dividing them according to the main challenge they address.

2) SIGNAL SPACE AMBIGUITY

The challenges in signal space ambiguity (detailed in Section IV-A3) lead to low floor distinguishability when using WiFi APs for multi-floor localization. The available solutions can be divided into (a) clustering-based approaches, (b) feature mapping, and (c) normalization approaches.

a: CLUSTERING

Clustering-based approaches focus on the effect that floor changes have on the signal in an isolated vertical cluster at the user's location, reducing overlap between floors in the signal space [51], [54], [60], [89], [98]. The challenge here is to estimate the correct cluster to be used for floor identification. For example, Shao et al. [54] present an adaptive algorithm for floor identification in challenging multi-floor environments using RSS and spatial similarity to cluster WiFi fingerprints based on augmented K-means

clustering, dividing the environment into 3D clusters containing similar RSS and spatial features. Bayesian models are then used to predict the probability distribution of clusters and floors, where each cluster has its own model for floor identification, and the output floor is estimated by maximizing the likelihood in the joint probability of the clusters and floors. Alitaleshi et al. [60] employ affinity propagation clustering and train a hierarchical extreme learning machine (ELM) to determine the cluster and floor based on WiFi signals. The 2D location is then obtained using a KNN approach based on the estimated cluster and floor.

b: FEATURE MAPPING

Some systems propose mapping the WiFi RSS into a different feature space that offers more floor distinguishability and better robustness to signal noise. A common approach is to employ autoencoders, such as stacked autoencoders (SAE) and denoising autoencoders (DAE), mapping the WiFi RSS into a latent feature space with better representation ability to identify the floor [52], [53], [62], [63], [64], [69], [73], [75], [78], [82], [83], [86], [96], [100]. These systems then apply the autoencoder to transform the input into this feature space before feeding it to the localization model, reducing the effect of signal space ambiguity. For instance, Kim et al. [82] train an SAE network and pass its output to a fully-connected classifier network to estimate the floor and the location of the user. JunLin et al. [75] use SAE with stacking model fusion of ELM, convolutional neural network (CNN), XGBoost, and support vector machine to perform floor identification. Yin et al. [64] propose JLGBMLoc in which they use a joint DAE to get a better fingerprint data representation.

c: NORMALIZATION

In order to handle the signal space ambiguity caused by the AP configuration and density, some systems include information on the AP configuration as part of the input, leveraging this additional information to improve floor distinguishability [47], [49]. For example, Elbakly et al. proposed TrueStory [47], which uses the installation floor of the APs as information regarding their deployment configuration. TrueStory filters RSS measurements to include only the RSS of APs within a window of floors around the floor of the AP with the strongest visible RSS, a process termed *vertical normalization*. The filtered RSS list is then fed to a group of weak learners to estimate the floor based on proximity-based hand-crafted features. To this end, TrueStory proposed a weak learner called the maximum transformed normalized average, which divides the sum of the RSS by the number of APs installed on the floor (not just those visible to the user), enhancing floor distinguishability in environments with challenging AP configuration and density. Similarly, StoryTeller [49] uses knowledge about the floors and 2D coordinates of the APs to perform *vertical* and *horizontal* normalization, increasing robustness to AP configuration. For horizontal normalization, StoryTeller only considers the APs in a certain horizontal area around the one with the strongest RSS.

3) LOW ACCURACY

To improve WiFi-based multi-floor localization accuracy, several systems have proposed utilizing recent DL models due to their remarkable modeling capabilities [49], [52], [53], [55], [61], [71], [72], [73], [75], [80], [87], [96], [100], [101], [104], [107]. In this section, we discuss three recent directions in which WiFi fingerprinting-based systems employ DL, divided into (a) RSS images, (b) RSS time series, and (c) deep reinforcement learning. Moreover, we outline the main limitations in DL-based approaches.

a: RSS IMAGES

Some systems leverage the image processing capabilities of CNNs with WiFi RSS fingerprinting, transforming RSS fingerprints into image representations and feeding them to a CNN-based model [49], [53], [71], [75], [80], [100], [107]. CNNs can detect intricate fingerprint patterns unique to the location and floor, improving accuracy. The main challenge becomes finding a meaningful image representation for the RSS fingerprints to enable the CNNs to learn these patterns. To this end, some systems proposed directly reshaping and padding the RSS fingerprint vector to generate a 2D image representation [80]. Although simple, this approach results in poor accuracy if several APs are unseen in each fingerprint vector, leading to a highly sparse image representation. To address this issue, some systems first perform AP selection before transforming the fingerprint vector into a 2D image, disregarding APs that have the least effect on the location prediction [53], [71], [100]. In [53], and [71], the authors rely on the Pearson correlation coefficient between the RSS of each AP and the fingerprinted locations when obtaining the 2D image representation, amplifying the features corresponding to the APs that are highly correlated with the fingerprinted locations. In [100], the authors perform dimensionality reduction on RSS fingerprint vectors using an ELM-autoencoder (AE) to obtain the salient features before reshaping them into a 2D image. In addition to this, using the RSS image representations allows capturing spatial information regarding the seen APs, further improving accuracy. For example, in [49], each AP is mapped to a certain pixel in the image according to its location and floor, allowing the system to consider this information when predicting the location.

b: RSS TIME SERIES

To accumulate more information, some systems proposed using a time series of RSS readings and employing a CNN [55], [73], [96], [101], [104] or a recurrent neural network (RNN) [52], [72], [87] to capture spatio-temporal patterns. This approach is more accurate than using a single RSS measurement due to accumulating information over time, providing more distinguishable fingerprints for each floor. For example, Ibrahim et al. [55] presented a hierarchical

architecture that builds a WiFi RSS time series from consecutive measurements and employs a CNN to extract temporal dependencies to predict building, floor, and location in a multi-building, multi-floor environment. ELesawi et al. [52] proposed multi-building and multi-floor localization using RNNs to estimate the building, followed by the floor and then the 2D location. The output of a higher level class (e.g., building) is passed to the lower level class (e.g., floor), leveraging the sequential nature of the RNNs. Note that using an RSS time series can significantly increase the localization latency depending on the scanning rate of the user device, due to relying on multiple consecutive measurements [124].

c: DEEP REINFORCEMENT LEARNING

In addition to the common supervised learning-based systems, reinforcement learning was used by Dou et al. [57] for multi-floor localization, achieving high accuracy. The authors formulate the multi-floor localization problem as a Markov decision process, and deep reinforcement learning is used to bisect the 3D search space of the multi-floor environment hierarchically, such that the floors of the environment restrict the possible vertical locations. This approach is more efficient than supervised learning-based approaches and can estimate the floor and location of the user simultaneously using a single model. Moreover, deep reinforcement learning allows learning after deployment to adapt to environmental dynamics that cause variations in WiFi RSS.

d: MAIN LIMITATIONS IN USING DL

Although DL-based approaches achieve high accuracy, their practical deployment is hindered by significant challenges. These include the need for extensive training data collection, re-building models for each new environment (i.e., low portability), and high storage and computational resource demands. In the following sections, we explore proposed solutions to address these limitations, dividing them into methods for reducing data collection overhead, improving portability, and minimizing resource requirements.

4) CALIBRATION DATA COLLECTION OVERHEAD

Solutions addressing data collection overhead can be divided into (a) crowdsourcing, (b) semi-supervised learning, (c) data augmentation, and (d) non-fingerprinting–based methods.

a: CROWDSOURCING

Crowdsourcing allows collecting large amounts of data without time-consuming manual site surveys by opportunistically leveraging additional sensors (e.g., barometers), when available, during the data collection phase to automate the location ground truth labeling process [7], [48], [56], [70], [94]. After data collection, the system switches to using only WiFi. For example, Shen et al. presented BarFi [48] for floor identification, which utilizes crowdsourced WiFi and barometer measurements, clustering these measurements to construct the WiFi fingerprint database. Ye et al. presented F-Loc [56] for floor identification, which uses

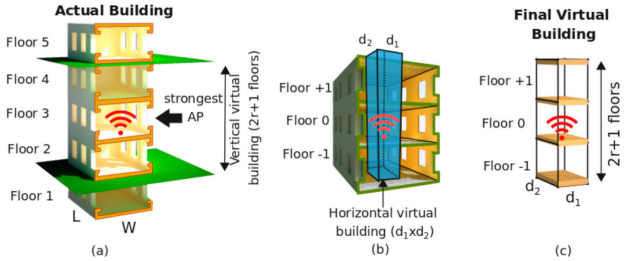


FIGURE 12. Virtual building construction process [49]. (a) The building is segmented vertically to include the floor with the strongest AP and r floors above and below it.(b) *Horizontal* segmentation defines a restricted area of $d_1 \times d_2$ around the strongest AP. (c) The final virtual building has an area $d_1 \times d_2$ and $2r + 1$ floors.

accelerometer readings to detect when users take an elevator and automatically start collecting WiFi signals at elevator entrances and exits. F-Loc then clusters the WiFi samples and orders them based on the floor level, identified with the help of accelerometer readings, to build the fingerprint database. Reliance on accelerometers instead of barometers in crowdsourcing improves the deployability of F-Loc, since accelerometers are more ubiquitous.

b: SEMI-SUPERVISED LEARNING

Semi-supervised learning assumes that only a subset of training data is labeled, leveraging the unlabeled data, which are easier to collect, to reduce data collection overhead [58], [70], [79], [102], [103], [105], [110]. This approach is often combined with crowdsourcing methods to eliminate the need to label all samples with additional sensors [103]. For example, Zhuo et al. proposed FIS-ONE [102], a system for floor identification using fingerprinting, which requires **only one** labeled sample collected on the ground or top floor. FIS-ONE employs an attention-based graph neural network to cluster WiFi signal samples and assign floor labels to the clusters. The number of clusters is set equal to the number of floors, and the system exploits the fact that signals from an AP are more likely to be detected on adjacent floors than on distant ones to order and label clusters effectively. Similarly, Shao et al. [70] introduced a semi-supervised Mean Teacher neural network model to label crowdsourced fingerprints and perform floor identification. The model uses transductive learning to mitigate the effects of time and device variability on fingerprints. Chen et al. [58] proposed a semi-supervised multi-kernel optimization learning scheme for floor and location estimation, requiring only 25% of the collected signals to be labeled. By mapping WiFi RSS data into high-dimension spaces using kernels, such as RBF and Chi-squared kernels, the authors obtain the location in a multi-floor environment. Wang et al. [110] proposed a WiFi fingerprinting-based system that employs semi-supervised learning and multi-task mean-teacher model to simultaneously predict the building, floor, and 2D location of the user. By leveraging multi-task learning, the system captures hierarchical spatial relationships more effectively

and improves generalization across buildings and floors. Thus, it achieves high multi-floor localization accuracy while significantly reducing the need for labeled data through semi-supervised learning.

c: DATA AUGMENTATION

Data augmentation creates synthetic samples from existing data, expanding the training data without additional collection overhead [70], [84], [99], [100]. For example, Shao et al. [70] proposed generating synthetic fingerprints at the midpoints of each three collected fingerprints by averaging their RSS, labeling the synthetic fingerprint with the center of gravity of the three original samples. Alitaleshi et al. [100] introduced a data augmentation technique that injects Gaussian noise into collected fingerprints to generate new samples. The variance of the injected noise for each AP is estimated according to the variance observed in the AP's fingerprint data. This approach simulates signal fluctuations caused by environmental factors, physical obstructions, and channel noise, enhancing the robustness to noise while generating realistic synthetic samples to augment training data.

d: NON-FINGERPRINTING–BASED

Some systems adopt zero-calibration approaches, leveraging WiFi infrastructure configurations and techniques that do not require data collection, such as proximity-based and ranging-based methods [47], [50], [68], [76], [77], [81], [91]. However, these systems typically deliver lower accuracy than fingerprinting-based approaches due to their sensitivity to infrastructure configuration and density [49]. For example, Bhargava et al. proposed Locus [50], a proximity-based system for multi-floor localization. Locus applies four proximity-based heuristics for floor identification: selecting the floor having the highest count of visible APs, selecting the floor of the AP with the highest RSS, selecting the floor with APs having the highest average RSS, and selecting the floor with APs having the highest RSS variance. Lohan et al. [91] presented a lateration-based approach using path-loss propagation models for multi-floor localization based on weighted centroids and non-negative least squares estimation. The authors evaluated two models: one without a floor-loss attenuation factor and another that incorporates a floor-loss factor. Their findings showed that the model with the floor-loss factor significantly improved floor estimation accuracy.

5) LOW PORTABILITY

Most WiFi-based systems face challenges with portability and generalization, often requiring the deployment process to be repeated for each new environment. To address this, Elbakly et al. proposed the StoryTeller system [49], which enhances portability by localizing users within a virtual building. This virtual building is composed of a constant window of floors, a fixed area, and relative AP locations instead of their absolute identifiers. The virtual building is constructed by extracting vertical and horizontal segments around the strongest visible WiFi AP (Figure 12). In addition, the system employs a fixed number of input features, represented as an RSS image with constant dimensions. Elbakly et al. validated StoryTeller's generalization capability by deploying it in unseen environments and applying transfer learning from the initial deployment instead of retraining the system entirely [49]. StoryTeller demonstrated robust performance with minimal fine-tuning data, significantly less than what was required during the original deployment. Recently, Narasimman et al. [109] proposed the fingerprinting-based Dumbloc system, which relies only on the single strongest WiFi AP and normalizes both the RSS input and floor label output, making the system independent of the environment. Dumbloc is shown to generalize well across unseen environments without significant fine-tuning.

6) RESOURCE REQUIREMENTS

Available solutions to enable the deployment of multi-floor localization systems on resource-constrained devices can be divided into (a) lightweight algorithms, (b) model compression, and (c) fingerprint shrinking.

a: LIGHTWEIGHT ALGORITHMS

Some systems employ lightweight models suitable for deployment on resource-limited devices. For example, ELMs require less computation and storage resources than common DL models. This is due to their shallow architecture and reliance on a single set of weights, making them easier to train and deploy [225]. ELMs have been successfully applied to multi-floor localization in studies such as [62], [63], [74], [85], and [97]. In addition, systems utilizing non-fingerprinting-based techniques [47], [50], [68], [76], [77], [81], [91] (Section V-A4d) often adopt lightweight proximity-based and ranging-based approaches. These methods are inherently less resource intensive, making them well suited for deployment on devices with limited resources.

b: MODEL COMPRESSION

Wang et al. introduced CHISEL [53], a multi-floor localization system designed for deployment on resource-constrained embedded systems. CHISEL integrates convolutional autoencoder (CAE), CNN, and model compression techniques, including quantization and pruning. Quantization reduces the bit width used to represent model weights and activations, while pruning removes less significant weights from the trained model. These approaches effectively lower the memory usage and computational requirements of the model. Moreover, CHISEL employs *quantization-aware* training, a technique that incorporates quantization effects during the training process, mitigating the accuracy loss associated with quantization-based compression.

c: FINGERPRINT SHRINKING

Several multi-floor localization systems aim to reduce the size of fingerprinting databases, allowing their deployment on resource-limited devices [62], [85], [88], [90], [92], [95], [106], [108]. A common approach involves dimensionality reduction techniques, such as linear discriminant analysis (LDA) or principal component analysis (PCA), to create a low-dimensional representation of signal fingerprints with minimal accuracy loss. For example, Luo et al. [106] use LDA to shrink the fingerprint database, obtaining the floor using a majority voting technique combined with a multi-classification one-versus-one decomposition method. Qi et al. [85] and Yan et al. [62] employed PCA with an ensemble of ELM models to achieve efficient multi-floor localization. Other systems leverage clustering to reduce the fingerprint database size by storing only the cluster representatives [88], [90], [92], [95]. Torres et al. [92] conducted a comprehensive analysis of different clustering methods, including k-means, affinity propagation, and rule-based, evaluating the trade-offs between loss of accuracy and reduction in storage and computational overhead. Cramariuc et al. [90] compared different clustering techniques based on parameters such as coordinates, 3D dimensions, APs, and fingerprinting reference points. They found that RSS-based clustering has lower complexity but worse accuracy compared to 3D-based clustering. In addition, feature selection has been employed by Ayinla et al. [108] on WiFi fingerprinting using recursive feature elimination with cross-validation to select only the most discriminative WiFi signal features before employing a deep neural network for inference. This approach ensures feature robustness, enhancing multi-floor localization, while reducing model complexity for deployment on resource-constrained devices.

B. BLUETOOTH-BASED SYSTEMS

1) BLUETOOTH IN MULTI-FLOOR LOCALIZATION

Bluetooth low energy (BLE) offers a communication range similar to classical Bluetooth but with significantly lower power consumption, making it an attractive choice for indoor localization systems. However, its adoption requires additional infrastructure deployment since typical indoor environments often lack the BLE beacon coverage needed for reliable signal transmission. This deployment cost becomes particularly significant in multi-floor environments, where beacons must be densely deployed on each floor to ensure robust multi-floor signal coverage. Moreover, BLE beacons require ongoing maintenance to address potential battery depletion or hardware failures [6].

A concern in BLE-based multi-floor localization is that BLE signals can be obstructed by dense building materials, floor slabs, or concrete structures, potentially degrading multi-floor localization performance. However, studies have shown that this issue can be mitigated by installing BLE beacons densely and strategically across floors to ensure adequate vertical signal coverage and diversity across floors. Specifically, several systems have successfully employed BLE for multi-floor localization by compensating for these structural limitations through planned beacon layouts and signal modeling [6], [111], [112], [113].

Compared to WiFi, BLE offers several advantages beyond lower energy consumption, including reduced transmission power and higher signal scanning rates on user devices, minimizing signal noise and multipath effects [226]. Moreover, BLE's shorter range makes it particularly suitable for environments with complex structures, such as atriums and hollow spaces between floors, as it helps confine signal propagation and improve floor discrimination [112]. Nevertheless, the wide availability of WiFi infrastructure makes WiFi a more common choice in practice. This section reviews systems that only depend on BLE to tackle the signal space ambiguity in challenging multi-floor environments [6], [111], [112], [113].

2) SIGNAL SPACE AMBIGUITY

Most systems that rely only on BLE are designed to perform well in environments with challenging structures, such as atriums and hollow spaces [6], [111], [112], [113]. For example, Qi et al. [112] proposed a fingerprinting-based adaptive weighted fusion algorithm for floor identification using BLE. The algorithm combines three components for floor identification: self-adaptive weighted BLE signal fusion, maximum average RSS, and an interval confidence-based approach. This method demonstrated robust performance in environments with atriums and hollow spaces. Hu et al. [6] developed the MLoc system, extensively evaluating it in different multi-floor malls. MLoc employs a proximity-based approach for floor identification, applying maximum voting among the floors of the five visible BLE beacons with the highest RSS. To address the signal space ambiguity challenges posed by atriums and hollow spaces, MLoc incorporates autoencoders to obtain a robust latent space, which is then processed by a fully connected classifier network. Recently, Wu et al. [113] presented a floor identification system that employs a floor-specific fingerprinting model using RSS measurements from strategically deployed BLE beacons, manually placed across floors to minimize signal overlap. The system trains a classifier for floor identification, ensuring reliable floor discrimination in complex multi-floor environments through optimized beacon deployment.

C. CELLULAR-BASED SYSTEMS

1) CELLULAR IN MULTI-FLOOR LOCALIZATION

Cellular networks consist of base stations, or cells, distributed over large areas to provide wide coverage. These cells range from macro cells, typically found outdoors, to indoor small cells (e.g., femtocells) designed to improve coverage within buildings [118]. The wide ubiquity of cellular technology, its high reliability and resilience in emergencies [124], and its minimal additional power consumption compared to the default operation of the user phone make it a promising option

for multi-floor localization. However, the large coverage of cellular base stations and the stability of their signals across different floors often leads to signal space ambiguity, resulting in lower multi-floor localization accuracy compared to other localization technologies (Section IV-A3). This section reviews systems that use only cellular for multi-floor localization [114], [115], [116], [117], [118], [119], [120], [121], [122], [123], [124], [125], [126], [127], categorizing them according to the primary challenge they address.

2) SIGNAL SPACE AMBIGUITY

The large signal coverage of cellular base stations leads to ambiguity in the signal space. Available solutions can be divided into (a) feature selection, (b) feature mapping, (c) sub-space–based estimation, and (d) indoor small cells.

a: FEATURE SELECTION

This approach selects the subset of cell towers whose signals have the highest floor distinguishability for accurate floor identification. For example, Varshavsky et al. proposed the SkyLoc system [114], which uses cellular RSS fingerprinting with KNN matching for floor identification. SkyLoc proposes per-floor feature selection by identifying the most effective towers to distinguish each floor. To do so, it filters out towers that are noisy or exhibit low RSS variability across floors, which is often the case for towers with large coverage areas. Zhou et al. [127] proposed IMPos, a multi-floor localization system based on 5G cellular fingerprinting. IMPos utilizes multibeam reference signal received power as the input fingerprint features and employs a stacked recurrent neural network to perform multi-floor localization. By exploiting the spatial selectivity and high resolution of 5G multibeam propagation, the system achieves high accuracy and overcomes the effect of signal space ambiguity where distinct beam patterns correspond to different spatial regions within the building.

b: FEATURE MAPPING

Some approaches map cellular signals into a feature space with enhanced floor distinguishability, reducing the effect of signal space ambiguity. For example, Rizk et al. presented CellRise [120], which uses deep LDA to map RSS vectors into a space with floor-distinguishing features, allowing accurate floor identification using RSS fingerprinting. CellRise then leverages a long short-term memory (LSTM) network to predict the floor from a sequence of RSS measurements.

c: SUB-SPACE–BASED ESTIMATION

Lv et al. [123] proposed IFLoc, a fingerprinting-based floor identification system that introduces subspace-based estimation to improve accuracy. IFLoc groups cellular RSS fingerprints into subspaces according to their horizontal locations, mitigating the problem of signal overlap between floors. A dedicated floor identification model is then trained for each subspace, improving floor distinguishability. To simplify the model selection process while maintaining high floor identification accuracy, IFLoc suggests using the serving cell tower associated with the user's device when grouping fingerprints. This eliminates the need to obtain the user's 2D horizontal position, which can be challenging in certain scenarios.

d: INDOOR SMALL CELLS

Since signal space ambiguity due to large signal coverage is generally in cellular-based systems that rely on outdoor cell towers, leveraging indoor small cells with their lower signal coverage can mitigate this challenge [117], [118]. However, small cell infrastructure is not as ubiquitous as common outdoor cell towers, and thus such systems would require additional infrastructure deployment costs.

3) CALIBRATION DATA COLLECTION OVERHEAD

Solutions for reducing the calibration data collection overhead in cellular-based systems can be grouped into (a) crowdsourcing, (b) weighted centroid, (c) data augmentation, and (d) non-fingerprinting–based methods.

a: CROWDSOURCING

Lv et al. introduced a novel crowdsourcing-based approach for data collection in their IFLoc floor identification system [123]. IFLoc uses GPS and barometer readings to label cellular RSS fingerprints with the ground truth floor. It begins by clustering barometer readings using DBSCAN to group similar pressure measurements. Then it estimates the vertical height of each cluster by referencing nearby outdoor readings of cellular, barometer, and GPS data. Using these outdoor measurements, IFLoc calculates height differences between floors, maps clustered barometer readings to their respective floors, and assigns corresponding labels to the cellular fingerprints collected alongside these barometer readings.

b: WEIGHTED CENTROID

Rizk et al. introduced the weighted centroid method in CellRise [120], omitting some floors during the data collection phase and determining the floor using the rounded value of the weighted centroid of the floor predictions. The weights are based on the floor probabilities estimated by the CellRise model. This approach requires data collection from only a subset of the environment's floors while still including the disregarded floors in the output predictions.

c: DATA AUGMENTATION

Mostafa et al. introduced UniCellular [124], [125], which employs various data augmentation methods to minimize the data collection overhead required for FCC-compliant floor identification. UniCellular integrates domain knowledge-based Gaussian noise injection with deep generative models based on a deep convolutional GAN (DCGAN) and a denoising diffusion implicit model (DDIM). This approach leverages diverse generative techniques to produce synthetic cellular fingerprints from varying data distributions,

significantly reducing the data collection overhead necessary to achieve accurate performance.

d: NON-FINGERPRINTING–BASED

Some systems rely on proximity-based and ranging-based approaches using indoor small cells [117], [118] to completely eliminate the data collection overhead. Rydén et al. [117] evaluated proximity-based cell identity and lateration-based TDoA methods for multi-floor localization using LTE small cells. The authors demonstrated that floor identification can be achieved with high accuracy when there is a sufficient deployment density of indoor small cells. These methods were further explored by del Peral-Rosado et al. [118], who conducted field tests using indoor femtocells for proximity-based and lateration-based floor identification.

4) LOW ACCURACY

To improve the accuracy of cellular-based multi-floor localization, recent systems have incorporated DL models, including CNNs [119], [124], [125], LSTMs [120], and autoencoders [122]. Despite these advancements, the accuracy remains relatively lower than that of WiFi-based approaches, particularly in environments with a limited number of visible towers [126]. To address this limitation, Mostafa et al. proposed CellFloor [126], which introduces a novel NLP-inspired method to represent sequences of cellular RSS measurements. CellFloor employs a transformer-based model for floor identification, capturing complex spatio-temporal patterns for distinguishing the floors. This approach allowed CellFloor to achieve near 100% floor identification accuracy, even in scenarios where only a single tower is visible.

D. BAROMETER-BASED SYSTEMS

1) BAROMETERS IN MULTI-FLOOR LOCALIZATION

Since atmospheric pressure decreases with increasing altitude, barometers have become a valuable tool for floor identification. Initially, barometers were large and impractical for field use [129], but are now widely available as embedded sensors and are becoming increasingly ubiquitous. Barometer-based systems typically follow one of two approaches for floor identification; they either detect changes in floors by analyzing variations in pressure readings and rely on a known initial floor to determine the current floor [129], [130] or determine the current floor directly by comparing barometer readings with absolute pressure values [128], [131], [132], [133], [134], [135], [136]. Here, we review systems that use the barometer only for floor identification [128], [129], [130], [131], [132], [133], [134], [135], [136], categorizing them according to the main challenge they address.

2) BAROMETER ENVIRONMENTAL SENSITIVITY

Environmental conditions (e.g., temperature) and device heterogeneity can make absolute barometer measurements unreliable for accurate floor identification (Section IV-A1). Solutions to address this challenge can be divided into using (a) reference stations, (b) reference barometers, and (c) crowdsourced calibration.

a: REFERENCE STATIONS

Some solutions rely on real-time pressure readings from nearby fixed reference stations, such as weather stations and airports, to calibrate barometer measurements [131], [135]. For example, Liu et al. [131] propose iBaro-altimeter, which integrates user barometer readings with air pressure data from nearby meteorological stations using an adaptive Kalman filter for floor identification. However, the sparse and infrequent updates from the reference stations pose a challenge in maintaining accuracy [131]. To overcome this limitation, Bai et al. [135] introduced a prediction algorithm using a Kalman filter to estimate current air pressure measurements at a reference station based on past measurements. This approach allows real-time calibration of barometer readings even when reference station broadcasts are infrequent, enabling accurate floor identification.

b: REFERENCE BAROMETERS

Another approach involves the installation of external reference barometers as local calibration points on different floors to provide real-time pressure data for floor identification based on user barometer readings [133], [134]. For instance, Xia et al. [133] proposed a solution in which multiple barometers are installed, one on each floor, and are used as references for user barometer measurements. Floor identification is performed by selecting the floor corresponding to the reference barometer with the closest pressure readings to the user barometer. Kim et al. [134] presented a method that requires a single reference barometer deployed on a floor with known altitude. Floor identification is achieved based on the height of the floors by comparing the pressure difference between the user's barometer and the reference barometer. Although this approach reduces infrastructure requirements by using only a single reference barometer, it depends on having accurate height measurements for each floor of the building.

c: CROWDSOURCED CALIBRATION

Crowdsourced pressure data can be used to calibrate barometers opportunistically based on user encounters [128], [136]. For example, Ye et al. presented SBC [136] and B-Loc [128], which build a fingerprint database of barometer measurements for floor identification and employ a crowdsourcing-based transitive calibration technique to achieve scalable barometer calibration. The core idea is to utilize user encounters to share and propagate pressure data, allowing continuous calibration among users. Each user then matches their real-time barometer readings with the fingerprint database to determine their floor. This approach mitigates the need for dedicated infrastructure (e.g.,

reference barometers), making it a scalable solution for barometer-based floor identification.

3) BAROMETER SENSOR DRIFT AND NOISE

Drift and noise in barometer measurements complicate the use of relative pressure differences to detect floor changes, since it becomes challenging to establish an accurate pressure difference threshold corresponding to a floor change. To address this, Li et al. [129] proposed a method to reset barometer errors using a nearby reference station before the errors accumulate substantially. Their approach first obtains the pressure reading on a known floor using the user barometer and determines the corresponding pressure reading from a nearby airport reference station. Then, the change in floor is estimated using the pressure changes at the reference station alongside those recorded by the user barometer. By combining the detected floor changes with a known initial floor, the system can accurately determine the current floor of the user.

E. INERTIAL SENSOR-BASED SYSTEMS

1) INERTIAL SENSORS IN MULTI-FLOOR LOCALIZATION

Inertial sensors, including accelerometers, gyroscopes, and magnetometers, are becoming increasingly available as triaxial sensors embedded in smartphones and other user devices. These sensors measure motion along the x, y, and z axes and are commonly used for PDR to track user movement. However, inertial sensors are prone to error accumulation caused by sensor noise, necessitating their fusion with other technologies. For example, inertial sensors are often combined with barometers to detect vertical motion activities, such as taking an elevator, since barometer measurements are more robust to variations in phone orientation and user activity compared to inertial sensors [130]. Some systems rely only on inertial sensors for infrastructure-free multi-floor location tracking [137], [138], [139], [140], [141], [142]. These systems typically assume that tracking is only required for short distances to avoid significant error accumulation. We classify these systems according to whether they require special hardware or operate using standard smartphone-embedded inertial sensors.

2) SPECIAL HARDWARE

Before embedded inertial sensors became ubiquitous in user devices, inertial sensor-based approaches often relied on wearable sensors [139], [140], [141]. Lee et al. [141] proposed using a body-worn sensing module for PDR in multi-floor environments by detecting three types of walking behaviors: walking on level ground, ascending stairs, and descending stairs. When stair movement is identified, the system updates the user's altitude by counting stair steps, allowing the determination of the vertical trajectory of the user. Similarly, Foxlin et al. introduced NavShoe [139], a system that employs shoe-mounted inertial sensors to track users in multi-floor environments. NavShoe also utilized a stair counting technique similar to the method of Lee et al. [141] to determine changes in altitude. Designed for mixed reality applications, NavShoe targeted short walking distances and recommended the use of floor plans and integration with other technologies to mitigate error accumulation for long-distance tracking.

3) NO SPECIAL HARDWARE (SMARTPHONE-BASED)

With the growing availability of inertial sensors in modern smartphones, such as accelerometers, gyroscopes, and magnetometers, several smartphone inertial sensors-based PDR systems have been developed for multi-floor localization without requiring any additional infrastructure or external hardware [137], [138], [142]. For example, Ye et al. presented FTrack [138], which uses a smartphone accelerometer for floor identification without requiring prior knowledge of the building or floor height. Leveraging user encounters, FTrack maps travel time (when using an elevator) or step counts (when using stairs) to the number of floors traversed, enabling the determination of the current floor. Wu et al. [137] proposed a PDR-based approach that uses smartphone inertial sensors for multi-floor location tracking. The system detects activities such as ascending or descending stairs and tracks height changes through stair step counting where the step length is set equal to a known stair step width. Chen et al. [142] proposed a smartphone inertial sensors-based floor identification system that analyzes pedestrian motion patterns in elevator and escalator scenarios and leverages an extended Kalman filter to estimate floor transitions.

F. HYBRID SYSTEMS

1) TECHNOLOGY FUSION IN MULTI-FLOOR LOCALIZATION

An increasingly popular approach to multi-floor localization involves combining multiple technologies. This approach addresses various challenges by leveraging the strengths of each technology to mitigate the limitations of relying on a single technology. Such hybrid approaches improve the availability and ubiquity of multi-floor localization systems by providing alternative localization methods tailored to different infrastructure and device capabilities. This section reviews different hybrid multi-floor localization systems [143], [144], [145], [146], [147], [148], [149], [150], [151], [152], [153], [154], [155], [156], [157], [158], [159], [160], [161], [162], [163], [164], [165], [166], [167], [167], [168], [169], [170], [171], [172], [173], [174], [175], [176], [177], [178], [179], categorizing them based on the technologies fused and highlighting the challenges each fusion resolves.

2) BAROMETER AND INERTIAL SENSORS

Some approaches combine barometers with inertial sensors [143], [144], [145], [146], [147], [148], [149], using a known initial location to perform PDR tracking. This fusion allows (a) improved landmark detection and (b) improved activity recognition, addressing input-level challenges in

using barometer and inertial sensor measurements (Section IV-A).

a: IMPROVED LANDMARK DETECTION

Detecting landmarks with known locations, such as specific stairs or elevators, allows resetting sensor errors during PDR. Barometers help identify vertical movements, while inertial sensors exhibit unique sensor patterns at these landmarks. When fused, these technologies improve the accuracy of landmark detection. For instance, Shang et al. proposed APFiLoc [146], which detects two types of landmarks: seed landmarks (e.g., stairs) and organic landmarks (e.g., anomalies in the magnetic field detected by magnetometers). Seed landmark locations are either fed to the system or inferred from the building's floor plan, while organic landmark locations are inferred from user trajectories in a crowdsourced manner. APFiLoc then employs barometers along with inertial sensors to identify landmarks, leveraging the reliability of relative pressure readings in detecting floor changes. A particle filter then fuses landmark data with sensor readings and floor maps, enabling multi-floor location tracking.

b: IMPROVED ACTIVITY RECOGNITION

Combining inertial sensors with barometers improves the accuracy of user activity recognition, providing opportunities to calibrate sensor noise and drift. For example, some systems use accelerometers to opportunistically calibrate relative barometer measurements for floor change detection [148], [149]. The core idea is to distinguish user mobility states, such as stationary, walking, and ascending or descending stairs, using accelerometer data and then adapt barometer readings accordingly. In addition, integrating barometers with inertial sensors improves the reliability of floor change detection compared to using only inertial sensors, as barometers are more robust to variations in user behavior and device orientation [130]. For instance, Shin et al. [145] proposed a multi-floor localization system where motion behavior is determined to select the appropriate PDR algorithm. Starting from a known location, height changes are estimated using relative barometer measurements to identify the current floor.

3) WIFI AND BAROMETER

Various approaches combine WiFi with barometers [150], [151], [152], [153], [154], [155], [156], [157], [158], [159], [160] to address input-level challenges related to WiFi signal space ambiguity and barometer environmental sensitivity, improving multi-floor localization accuracy by enabling (a) infrastructure-free floor change detection and (b) crowdsourcing and automatic barometer calibration.

a: INFRASTRUCTURE-FREE FLOOR CHANGE DETECTION

The signal space ambiguity inherent in using WiFi for multi-floor localization (Section IV-A3) can be mitigated by integrating infrastructure-free relative barometer measurements to detect floor changes. Barometer readings help resolve the ambiguity caused by relatively stable WiFi signals across different floors, particularly in environments with challenging structures (e.g., hollow spaces). Moreover, barometers improve multi-floor localization accuracy in the presence of multipath effects and signal interference that can affect WiFi signals. For example, Huang et al. [152] combined WiFi and barometer data to achieve accurate floor identification in complex environments that contain atriums and large hollow spaces. Their system utilizes a hidden Markov model and XGBoost to fuse these two data sources.

b: CROWDSOURCING AND AUTOMATIC BAROMETER CALIBRATION

The unreliability of absolute barometer measurements and the effect of barometer sensor noise can be mitigated by leveraging WiFi for absolute localization and automatic calibration of barometer readings during floor change detection. Moreover, barometer readings can be utilized in crowdsourcing to automatically label WiFi fingerprint data based on pressure changes. For example, Zhao et al. proposed HYFI [153], which integrates WiFi AP distribution with barometer readings using Bayesian classification for floor identification. Floor information from the AP distribution is used to initialize and calibrate barometer readings. Similarly, Elbakly et al. introduced HyRise [150], which combines WiFi fingerprinting with barometer measurements using a Markov localization framework for floor identification. HyRise crowdsources barometer readings to calculate pressure differences between floors and builds a histogram representing the installation floors of each WiFi AP. The Markov framework updates the probability distribution of the user's floor by incorporating barometer-detected floor changes and WiFi RSS observations based on the constructed AP histogram. HyRise accommodates devices that lack barometers by allowing WiFi-only localization, allowing broader applicability.

4) WIFI AND INERTIAL SENSORS

Due to the higher ubiquity of inertial sensors compared to barometers [166], some systems have proposed combining WiFi with inertial sensors instead of barometers [161], [162], [163], [164], [165], [166]. Similar to the fusion of WiFi and barometers, this integration addresses the challenges associated with WiFi signal space ambiguity and inertial sensor noise and drift, offering (a) infrastructure-free floor change detection and (b) crowdsourcing and automatic inertial sensor calibration.

a: INFRASTRUCTURE-FREE FLOOR CHANGE DETECTION

Inertial sensors help address challenges related to the WiFi infrastructure that can cause ambiguity in the WiFi signal space (Section IV-A3). Infrastructure-free inertial sensor readings enable floor change detection by identifying events such as taking stairs or elevators, avoiding signal space ambiguity issues. For example, Sun et al. [161] proposed

a system for multi-floor localization in environments with challenging structures, such as atriums, by using WiFi fingerprints combined with a floor discriminative model trained using LDA. This approach maximizes the distinction between neighboring floors in the signal space. To ensure reliable multi-floor localization in complex environments, the floor is updated only when inertial sensors detect a floor change.

b: CROWDSOURCING AND AUTOMATIC INERTIAL SENSOR CALIBRATION

WiFi-based localization can provide an initial location and reset error accumulation when integrated with inertial sensors for PDR by estimating the absolute location. Furthermore, inertial sensor measurements provide continuous walking trajectories and floor change detection, enabling automatic labeling of crowdsourced WiFi readings. For instance, Radu et al. proposed the HiMLoc system [164], which combines activity recognition and PDR from smartphone inertial sensors with WiFi fingerprints using a particle filter for multi-floor localization. HiMLoc uses collected WiFi fingerprints and incorporates landmark detection based on the building structure to counteract accumulation of inertial sensor errors.

5) WIFI, BAROMETER, AND INERTIAL SENSORS

The fusion of WiFi, barometers, and inertial sensors for multi-floor localization is a widely adopted approach [167], [167], [168], [169], [170], [171], [172], [173], [174], [175], [176], [177], [178], [179], as it combines the benefits of each technology to further improve accuracy and availability. In this fusion, WiFi is used for absolute localization, providing an initial position and resetting accumulated errors, while barometers, combined with inertial sensors, are responsible for height change detection and continuous location tracking through PDR. For example, Yu et al. [168] proposed a system that integrates barometer- and inertial sensor-based PDR with WiFi fingerprints for multi-floor localization. It uses an adaptive particle filter to fuse WiFi fingerprint matching results, obtained through a WKNN, with sensor measurements to estimate the location. Zhang et al. [171] formulated fusion as a recursive function approximation problem. Their system processes a time series of WiFi and inertial sensor measurements using an LSTM network to track the location of the user. For floor identification, the system combines barometer readings with the WiFi AP distribution through Bayesian classification, similar to the HYFI system [153]. Yu et al. [176] introduced a PDR trajectory optimization framework that combines WiFi finite-time measurement anchors with inertial and barometer sensors. The system employs an unscented Kalman filter to fuse WiFi and sensor readings. It combines lateration based on WiFi signal RTT and RSS information with landmark detection through dynamic time warping, providing absolute references for resetting PDR errors.

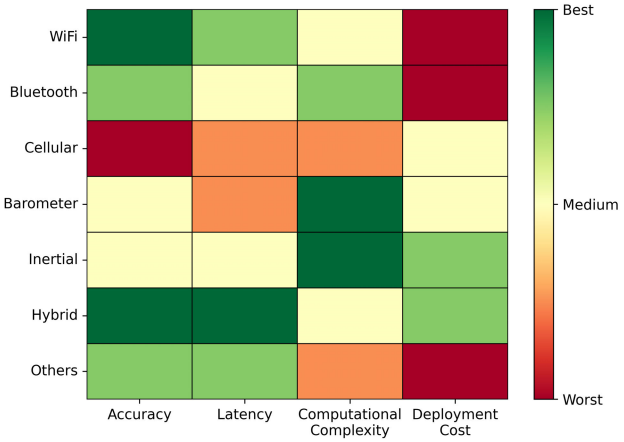


FIGURE 13. Summary of the trade-offs across technologies used in multi-floor localization.

G. OTHER TECHNOLOGIES

Technologies other than those discussed are less commonly used in multi-floor localization systems [180], [181], [182], [183], [184], [185], [186], [187], [188], [189], [190], [191], [192], [193], [194], [195], [196], [197], [198], [199], [200], [201], [202], [203], [204], [205], [206], [207], [208]. These systems face certain technology-specific limitations that restrict their widespread adoption, as they either have limited transmission range [188], [189], [190], [191], [192], [193], [194], [195], [196], [197], [198], [199], [200], [201], [202], [203], [204], [205], [206], [207], require additional infrastructure costs [180], [184], [186], [187], [208], have high power consumption demands [181], [185], or provide relatively low accuracy [182], [183]. However, these systems can be suitable for specialized use cases where their specific advantages outweigh the limitations.

1) LIMITED BY TRANSMISSION RANGE

While technologies such as ultra-wideband (UWB) [188], [189], [190], [191], [192], [193], [194], radio frequency identification (RFID) [195], [196], [197], [198], [199], [200], [201], [202], and acoustic signals [203], [204], [205], [206], [207] have demonstrated high precision in fine-grained 3D indoor localization in small areas, e.g., achieving decimeter-level accuracy in the case of UWB [188], their use in multi-floor localization systems remains limited in practice. This is because these technologies are generally constrained by significantly short transmission ranges [227], requiring high deployment cost overhead to provide the needed density of signal transmitters (UWB anchors, RFID active tags and readers, or audio beacons) to cover multi-floor environments [189], [190], reducing their suitability for multi-floor localization. Nevertheless, with sufficient transmitter infrastructure deployment, these technologies can provide accurate multi-floor localization. For example, Ajroud et al. [195] proposed a multi-floor localization system where they deployed **multiple** RFID readers across

TABLE 4. Performance trade-offs in multi-floor localization systems. Metrics are either reported in the reviewed papers or estimated based on technology characteristics and employed methods.

Technology	Accuracy	Latency	Computational Complexity	Deployment Cost	Notes
WiFi [7], [49]–[112]	Exact floor estimation accuracy: 90-100%; avg. error: 6.5 m	Low-Medium	Medium	Medium-High	Widely used. Offers high accuracy with deep learning (e.g., CNN [57]), but requires medium latency for sequence capture and incurs high deployment cost due to fingerprinting overhead.
Bluetooth [6], [113]–[115]	Exact floor estimation accuracy in complex environments with atriums and hollow spaces: 93-96%	Medium	Low-Medium	Medium-High	Less common than WiFi. Can achieve high accuracy in complex environments (e.g., with atriums), but incurs high deployment cost due to beacon infrastructure.
Cellular [116]–[129]	Exact floor estimation accuracy: 73-99.9%	Low-High	Low-High	Medium-High	Less common than WiFi. Accuracy ranges from low (e.g., KNN [116]) to high with complex models (e.g., LLMs [128]), requiring high latency and deployment cost.
Barometer [130]–[138]	Exact floor estimation accuracy: 98%; avg. vertical positioning error: 0.5 m in **short-term tracking**	Medium-High	Low-Medium	Medium-High	Barometer-only systems are uncommon due to environmental sensitivity, requiring calibration and resulting in high latency, complexity, and infrastructure deployment cost.
Inertial Sensors [139]–[144]	Exact floor estimation accuracy: 97% in **short-term tracking**	Low-High	Low	Low-High	Rarely used alone due to error accumulation. High accuracy requires either high latency or special body-worn hardware, and performance degrades over long trajectories.
Hybrid [145]–[182]	Exact floor estimation accuracy: 87-100%; avg. error: 1.4 m	Low-Medium	Low-Medium	Low-Medium	Commonly used. Combines complementary technologies for high accuracy with low latency, complexity, and deployment cost—but depends on availability of specific sensor combinations on user devices.
Other Technologies [183]–[211]	Exact floor estimation accuracy: from 90%; avg. vertical positioning error: down to 5 cm	Low-Medium	Low-High	Medium-High	Less common due to technology-specific constraints (e.g., high power consumption [184], [188]). High accuracy often requires increased complexity or deployment cost, with latency varying by method.

floors, allowing for precise floor-specific tracking using 3D holographic representations from RFID signal phase data. By optimizing RFID reader placements, their system achieved an error of only 5 cm in a multi-floor environment. Thus, while these technologies offer strong potential for fine-grained indoor localization, their limited transmission range necessitate high deployment costs for practical use in multi-floor localization.

2) LIMITED BY INFRASTRUCTURE COSTS

Some systems require additional infrastructure deployment costs, including, e.g., those that use surveillance cameras [184] or long range (LoRa) technology [208], whose infrastructure is often unavailable in typical environments. For instance, Yan et al. [184] presented a system that fuses multiple surveillance cameras with smartphone-based PDR for multi-floor localization. The system tracks motion using a DL-based approach for real-time detection using surveillance cameras and smartphone-embedded barometers to detect floors and transition areas. The 3D location of the cameras helps in floor detection. Chen et al. [208] proposed deploying LoRa nodes for fingerprinting-based multi-floor localization, achieving high accuracy. Their approach combines time-based ranging and RSS with Gaussian filtering for noise reduction, followed by a random forest classifier algorithm for floor identification. Although these approaches

can provide high accuracy, their significant infrastructure deployment and maintenance costs make them less common in multi-floor localization.

3) LIMITED BY HIGH POWER CONSUMPTION

Some systems integrate smartphone cameras, which provide images for visual positioning information, with PDR [181], [185]. However, due to the high power consumption of cameras, such systems quickly drain the phone battery, limiting their usability for prolonged periods. For instance, Qian et al. [181] introduced a system that combines inertial sensors with images from the smartphone camera. The system uses CNN networks to match real-time images with location-tagged images collected offline, identifying the location corresponding to the real-time image. This provides an absolute reference for correcting PDR errors. By collecting location-tagged images on each floor, visual positioning enables the system to provide significantly high accuracy in multi-floor environments.

4) LIMITED BY RELATIVELY LOW ACCURACY

Some systems utilize infrastructure-free magnetic [182], [183], [228] or frequency-modulated (FM) radio signal fingerprints [229] to provide multi-floor localization, commonly fused with smartphone inertial sensor-based PDR. However, these fingerprints offer lower localization accuracy compared to other technologies (e.g., WiFi) due to their significantly lower fingerprint granularity, where magnetism provides a single fingerprint dimension corresponding to the magnetic field strength [230] and FM signals come from few distant broadcast towers with low inter-floor signal differentiation. For example, Zhao et al. [182] proposed a system that integrates magnetic fingerprints and inertial sensor readings for floor identification. The system employs inertial sensors for activity recognition, such as normal walking and shaking of the phone. Based on the detected activity, real-time magnetic data are mapped to the appropriate fingerprints, and KNN is used for fingerprint matching. Liu et al. [228] proposed a multi-floor navigation system based on magnetic fingerprints using an optimized ant colony algorithm, exploiting floor plans and landmarks. Cong et al. [229] designed a multi-floor localization system using FM radio signals and smartphone inertial sensors where features are selected to allow recognizing activities such as level walking, ascending, and descending stairs. A random forest classifier combines the features to estimate the user's floor. Although these systems require no infrastructure costs, their accuracy is relatively limited compared to, e.g., WiFi and inertial sensor fusion.

H. TECHNOLOGY TRADE-OFFS AND SUITABILITY

To study the trade-offs across the technologies used in multi-floor localization, we provide a comparative summary of the reviewed systems based on four key performance metrics: *accuracy, latency, computational complexity, and deployment cost*. These trade-offs must be carefully studied and managed when making system design and deployment decisions [231], [232]. The full details of our comparative analysis on example multi-floor localization systems can be found in Table 5 in Appendix A, with Table 4 and Figure 13 summarizing the trade-offs among the technologies for quick reference. The main conclusions based on our comparative analysis can be summarized as follows.

- **WiFi** is the most commonly used wireless technology in multi-floor localization as it offers relatively high accuracy (up to 100% exact floor estimation accuracy using deep learning models and a sequence of samples [55]) while relying on the ubiquitous deployment of WiFi APs. However, WiFi-based systems typically rely on fingerprinting, which results in high deployment costs due to the fingerprinting data collection overhead.
- **Bluetooth** can achieve accuracy levels comparable to WiFi, especially in environments with hollow spaces and atriums, where its shorter transmission range limits cross-floor interference. However, Bluetooth signals are more susceptible to blockage by thick floors or dense construction materials, which can degrade performance in multi-floor structures. To overcome this, high-accuracy Bluetooth-based multi-floor localization requires dense beacon deployment to ensure reliable signal availability on each floor [113]. This leads to increased deployment and maintenance costs, limiting the widespread adoption of Bluetooth for multi-floor localization.
- **Cellular** solutions rely on ubiquitous highly stable infrastructure, making them suitable for safety-critical applications such as emergency response [125], but require highly complex models (e.g., LLMs [126]) and a long sequence of samples to reach an accuracy comparable to WiFi-based solutions, increasing their latency and complexity. This makes them less favorable for lightweight real-time multi-floor localization applications.
- **Barometer and inertial** sensors, while limited in estimating the absolute user floor when used alone, offer highly accurate floor change detection with low latency and complexity when used for short user trajectories. Furthermore, systems based on these technologies typically rely on lightweight algorithms and are infrastructure-free, making them attractive for resource-constrained devices where power is limited, while requiring low deployment costs. However, using barometer and inertial sensors without fusion with other technologies is uncommon due to the effect of environmental sensitivity on barometer readings and the accumulation of errors caused by sensor drift and noise.
- **Hybrid solutions** strike a promising balance across all four metrics, achieving high accuracy with relatively moderate latency, complexity, and deployment cost.

However, their applicability is bounded by the availability of all employed sensors in user devices.

- **Other technologies** are less common due to technology-specific constraints. For example, systems that rely on smartphone cameras to obtain visual information can achieve high accuracy, but have high power consumption requirements, increasing their complexity [181], [185]. Magnetism-based systems are infrastructure-free, resulting in relatively low complexity and deployment costs, but provide relatively low accuracy (e.g, compared to WiFi-based solutions) [182].

Overall, the suitability of a technology for multi-floor localization depends on *specific application constraints*, e.g., whether accuracy is the top priority, or if the system must operate under strict latency, deployment cost, or resource limitations. For example, if accuracy is the top priority, then deep learning-based systems that leverage WiFi or a hybrid of WiFi, barometer, and inertial sensors are the most reliable [233]. If low latency infrastructure-free floor change detection for short-term tracking is desirable, then barometer and inertial sensor-based systems are best [136], [138]. If persistence of the system in emergencies is required, then cellular-based systems that rely on outdoor base stations are favorable [124].

Furthermore, given the increasing importance of indoor positioning for IoT applications, where devices are often resource-constrained, battery-powered, and latency-sensitive, computational complexity can be a critical factor for many applications. Techniques requiring deep models, long sequences, or extensive calibration (e.g., full fingerprinting) may be unsuitable for such cases. On the other hand, systems optimized for lightweight modeling [50], [53], [54] or technology fusion [146], [148], [161] to reduce the needed computational complexity can be a suitable middle ground for practical IoT deployments.

Our comparative analysis of the technology trade-offs in Table 4 and Figure 13 offers a high-level guide for choosing the most appropriate technology for a given use case.

VI. OPEN PROBLEMS AND FUTURE DIRECTIONS

This section presents several open problems and new directions for future multi-floor localization research.

A. ENVIRONMENT-RELATED DIRECTIONS

1) UNIFIED BENCHMARKING PLATFORMS

Despite the progress in multi-floor localization research, evaluating system performance remains highly fragmented due to inconsistent environment testbeds, sensor configurations, and reporting standards. This fragmentation hinders the development of systems capable of operating reliably in large-scale, real-world deployments involving tall and multi-building environments; settings increasingly common in urban centers such as residential compounds, multi-building campuses and shopping malls, and skyscrapers [234], [235].

Most existing systems are evaluated in isolated, single multi-floor buildings, limiting their generalizability in multi-building environments. In such scenarios, a system must first identify the correct building before estimating the floor and 2D location. Hierarchical classification approaches have been proposed to handle this added complexity, where a separate model is trained for each building and floor [52], [55], [59], [69], [72], [73], [74], [82], [104]. However, this strategy faces poor scalability due to the exponential increase in model count with larger deployments. An alternative solution involves training a single, unified model to simultaneously estimate the building, floor, and 2D location [61], [67], [71], [78], [80], [83], [86]. Although this approach improves scalability, it often comes at the cost of reduced accuracy and increased model retraining complexity when environmental structure changes occur. A promising hybrid direction is to cluster adjacent floors within each building and assign a shared model to each cluster, reducing the number of required models while maintaining an acceptable scalability and accuracy trade-off.

Taller buildings further exacerbate these challenges. Although there are more than four million high-rise buildings [234] and thousands of skyscrapers [235] worldwide, most studies focus on mid-rise buildings with fewer than ten floors [6], [119], [150]. Taller structures increase signal space ambiguity across floors (Section IV-A3), require denser transmitter deployments for fingerprinting-based systems, and degrade barometer and inertial tracking accuracy due to cumulative sensor errors (Section IV-A2). Hybrid systems that combine multiple technologies (Section V-F) or leverage optimizied placement of signal transmitters, such as WiFi APs [236], [237], may alleviate some of these constraints.

To systematically evaluate these trade-offs and ensure reproducibility, there is an urgent need for a unified benchmarking platform. Such a platform should include standardized, publicly available datasets with floor-level annotations across multiple buildings and high-rise structures, complete with floor plans and sensor logs. It should also define evaluation protocols for both horizontal and vertical errors (in meters and floors), latency, and deployment scalability. Such a benchmarking platform would enable head-to-head comparisons between techniques under realistic deployment scenarios, leading to more robust and generalizable solutions for real-world scenarios such as multi-building and tall-building environments.

2) FLOOR TRANSITION DETECTION IN CHALLENGING STRUCTURES

Accurately detecting floor transitions remains a critical but underexplored challenge in multi-floor localization, particularly in buildings with non-standard structural layouts. Existing infrastructure-free approaches that rely on inertial sensors [137], [142] or barometers [134], [135] often assume ideal conditions, such as conventional staircases and uniform

floor heights; however, these assumptions do not hold in many real-world environments.

Inertial sensor-based methods typically estimate the number of floors traversed by detecting staircase landings during ascent or descent [143]. Thus, these systems face significant limitations in the presence of unconventional stair types such as straight, spiral, or circular staircases that lack intermediate landings. Moreover, environments like airports and shopping malls commonly use ramps and escalators, which produce smooth vertical movement without distinct step patterns, making conventional inertial sensor-based step detection techniques ineffective [142]. To address these limitations, future systems must be able to recognize a wider range of vertical movement patterns. Promising directions include training machine learning models on diverse vertical transition types, e.g., unconventional stairs, ramps, and escalators, using multimodal data from inertial sensors, magnetometers (e.g., to detect escalator motors), and activity recognition. Developing classifiers tailored to detect diverse types of movement will allow floor change detection systems to generalize more effectively to complex environments.

Barometer-based systems, on the other hand, commonly assume constant pressure differentials between adjacent floors [134], [135]. This assumption fails in buildings with uneven floor heights, such as split-level buildings, large theaters, or stadiums [143], where the vertical spacing between floors varies, making barometer-based floor transition detection unreliable. To overcome this, barometer-based floor change detection algorithms should incorporate adaptive floor height models, either through sensor fusion (e.g., with inertial data) or probabilistic frameworks such as Bayesian inference using floor height distributions as priors.

Overall, the need for robust and scalable floor transition detection is especially crucial in real-world deployments across structurally diverse buildings. Future research should explore lightweight, infrastructure-free approaches that combine activity recognition, sensor fusion, probabilistic models, and semantic context (e.g., detecting transitions near escalators). To support these advancements, standardized testbeds with labeled floor transitions and varied structural features are essential, enabling reproducible evaluation and fair comparison across methods and accelerating the development of universally applicable floor transition detection approaches.

3) ROBUSTNESS TO DYNAMIC ENVIRONMENTAL CHANGES

Multi-floor localization systems deployed in real environments must remain robust against dynamic and unpredictable conditions. However, most current evaluations occur on static testbeds and overlook the effects of human presence, elevator movement, and structural changes, such as updated floor plans or relocated signal transmitters.

For example, human presence and physical obstructions (e.g., furniture or closed doors) in hallways or rooms can lead to signal occlusion, multi-path effects, and transient signal strength fluctuations, particularly affecting fingerprinting systems by distorting the received signal patterns compared to the stored fingerprinting database [44], [162]. Systems should be designed to handle such disturbances, using techniques such as real-time noise filtering mechanisms, ensemble predictions, context-aware outlier detection, or fusion with other sensing modalities (e.g., combining inertial and barometric data) to smooth out anomalies. Moreover, adopting more robust signal features can improve resilience to temporary signal fluctuations caused by user movement or environmental variations. For example, fingerprinting the RTT, which has shown improved stability in 2D positioning [9]. Furthermore, floor plan updates (e.g., room layout changes, wall additions) and transmitter reconfiguration can cause fingerprinting methods to fail unless continuously adapted. Incremental or transfer learning and adaptive model retraining must be incorporated into future systems to maintain accuracy in case of such long-term environmental changes.

Similarly, elevator movement dynamics present unique challenges for barometer and inertial sensor-based floor transition detection. Unlike stair climbing, where step patterns and elevation changes can be clearly detected, elevator acceleration and deceleration profiles vary depending on the number of floors traveled, the load weight, and the elevator control logic. These variations introduce unpredictable inertial signals, while barometric pressure changes can be inconsistent due to sealed shafts or airflow damping. To overcome this, sensor fusion techniques can be applied, using models trained on elevator-specific motion profiles. For example, these models can detect elevator usage by identifying periods of accelerometer stillness combined with sudden pressure drops or rises, tracking vertical position changes.

Evaluating robustness to such dynamics requires benchmarks with annotated dynamic scenarios and simulation tools that model changing environments. Creating such resources will be essential for designing and testing robust and adaptable systems for real-world deployments.

4) REAL-WORLD DEPLOYMENT EVALUATION METRICS

Most existing multi-floor localization studies emphasize offline accuracy metrics, often neglecting the challenges associated with real-time deployment. This fails to capture key operational constraints, such as responsiveness to environmental structure variability and reliability in safety-critical scenarios. Future research must prioritize real-time metrics that capture how quickly and reliably a system can respond to such changes.

For instance, floor-based outputs are often ambiguous in buildings with non-uniform floor numbering schemes (e.g., skipped numbers or localized labeling) and in emergency scenarios where floors may collapse or be mislabeled [2]. Future systems should instead output absolute height estimates (e.g., meters above ground level) alongside or in place of floor numbers. This approach provides a more reliable reference, especially in emergency response situations.

Moreover, existing systems rarely provide an indication of the confidence level of their predictions, which is essential for risk-sensitive decisions [2]. Incorporating uncertainty estimates, such as confidence intervals or class probabilities, would allow systems to communicate prediction reliability, enabling users to interpret localization results with greater caution [2].

In general, research should move toward real-time evaluation frameworks that assess performance under dynamic and evolving real-world conditions [6]. Standard metrics must expand beyond offline localization accuracy to include time-to-fix, adaptation time to environment changes, and location confidence. Developing a standard real-time evaluation framework that includes these metrics would enhance benchmarking across systems and ensure deployment-readiness beyond simple offline accuracy.

B. DEPLOYMENT OVERHEAD-RELATED DIRECTIONS

The overhead of collecting labeled training data in multi-floor environments is substantial, as each floor requires extensive site surveys (Section IV-B1). Although existing solutions have mitigated this challenge to some extent (Section V), there is significant room for further improvement.

1) UNSUPERVISED MULTI-FLOOR LOCALIZATION

One common method to reduce data collection overhead involves crowdsourcing and leveraging additional sensors to automatically label fingerprints [48], [94]. However, crowdsourcing with additional sensors for labeling requires a sufficient number of devices equipped with these sensors to collect the data. Moreover, the process of curating and cleaning crowdsourced data is labor-intensive and prone to labeling errors, which can compromise the accuracy of the resulting dataset. Thus, some systems employed semi-supervised learning to use a small number of labeled samples, manually tagged with ground truth to ensure accurate labeling, to label the remaining data [102], [103]. However, manually tagging samples relies on active user participation, which can be unreliable without sufficient incentives and raises privacy concerns. A promising direction is to develop fully unsupervised learning-based systems to label crowdsourced fingerprints without any labeled samples. Such systems can utilize building layouts (e.g., number of floors and infrastructure configuration) and clustering algorithms to automate labeling. Recently, FIS-ONE [102] demonstrated the feasibility of labeling crowdsourced samples for floor identification using **only one** labeled sample from the ground floor, paving the way for a fully unsupervised fingerprinting-based floor identification approach.

2) AUTOMATIC CONSTRUCTION OF FLOOR DATA

Some methods proposed generating data automatically using generative models and data augmentation techniques [124], [125]. However, these methods still require a few labeled samples per floor to learn the data distribution. A promising direction is to develop approaches to generate data for **unseen** floors based on data from **adjacent** floors. Extending spatial augmentation techniques used in 2D localization, where data from surrounding 2D locations are used to generate fingerprints for unseen areas [23], could achieve this goal. This would significantly reduce data collection requirements for multi-floor environments.

3) TRANSFERRING CALIBRATION DATA ACROSS NEARBY BUILDINGS

Current systems require collecting data from all buildings and floors in multi-building environments, leading to significant deployment overhead. Many clustered buildings have similar heights, structures, and signal distributions across floors with the same height [3]. Studying the transferability of calibration data across nearby buildings can be a promising direction to reduce the deployment data collection overhead. For instance, data can be collected from the floors of one building and adapted to nearby buildings, accounting for differences in distance and layout. This strategy is particularly promising for cellular-based systems, as nearby buildings are often served by the same cell towers, resulting in similar signal distributions [126]. Similarly, barometer-based approaches can benefit from the consistent air pressure and relative pressure differences observed across adjacent buildings. Moreover, WiFi-based systems can apply this method if the WiFi AP deployment configurations are similar across nearby buildings. Such a strategy has the potential to significantly reduce deployment overhead in clustered multi-building environments.

C. USER AND DEVICE-RELATED DIRECTIONS

1) ROBUST PDR

Systems using inertial sensors for PDR and floor transition detection, such as [137], [164], [169], [172], and [177], must adapt to user-specific characteristics and activities to maintain accuracy. Current methods rely on user-specific calibration that limits their generalizability.

a: USER CHARACTERISTICS

Variations in user walking speeds can cause up to $\pm 50\%$ differences in step lengths [8]. Moreover, user characteristics, such as gender, weight, height, and age, further contribute to differences in step length. Even the same user may frequently change their walking speed, for example, when navigating through crowded areas. These variations must be taken into account in PDR step counting and floor change detection, particularly when estimating the step length. Existing techniques either assume a constant, user-calibrated step length based on their average walking speed or require user-specific calibration parameters to estimate step lengths adaptively using accelerometers [238]. A promising direction is to design an adaptive step length estimation technique that eliminates the need for user-specific calibration and considers vertical motion for multi-floor localization. Such a method should dynamically adjust to variations in walking speed,

especially during activities such as climbing stairs, and apply the appropriate step length accordingly.

b: USER ACTIVITY AND DEVICE PLACEMENT

User activities, such as making phone calls, swinging of the arm, or texting, along with changes in device placement, such as in-pocket vs in-hand, introduce significant variations in inertial sensor readings. These variations complicate PDR step counting and often lead to false floor change detections in multi-floor environments. Current systems address this challenge by identifying user activities and device placements from a predefined set and adapting their PDR algorithms accordingly [137], [164], [177]. To improve robustness and generalization, multi-floor PDR systems should function reliably with any arbitrary user activities and device placements [239]. One potential approach is to use unsupervised domain adaptation ML techniques, enabling the system to learn a latent space that is invariant to changes in inertial sensor data caused by user activity or device placement [240].

2) DEVICE HETEROGENEITY

Heterogeneous hardware and software specifications between devices, even between identical models [8], result in distinct sensor measurements at the same location and time. This inconsistency increases localization ambiguity in multi-floor environments. Since it is impractical to train systems with data from every possible device, training data are typically collected from a limited set of devices. Consequently, systems trained on these data often perform poorly when deployed on unseen devices [24]. Unfortunately, current multi-floor localization systems largely ignore this challenge. Promising directions to address device heterogeneity include leveraging relative measurements [241], which are less dependent on absolute sensor values, and learning device-invariant representations using autoencoders [24], which have demonstrated robust performance in 2D localization settings. Adapting these solutions for multi-floor localization and evaluating their performance in multi-floor environments can enhance robustness across diverse devices.

D. SECURITY AND PRIVACY-RELATED DIRECTIONS

While some multi-floor localization systems rely on lightweight models deployed directly on the user device [53], most systems offload heavy computations to a server to accommodate the resource-intensive nature of multi-floor localization systems (Section IV-B3). Moreover, systems that reduce data collection overhead through crowdsourcing (Section IV-B1) require users to transmit sensitive location data to a server [48], [94]. These practices raise critical concerns regarding security and privacy.

1) SECURITY

Outsourcing computations to a server introduces vulnerabilities, such as attacks on the communication between user devices and the server. For example, distributed denial-of-service (DDoS) attacks can disrupt localization services by preventing user devices from uploading data or receiving location estimates from the server [242]. Implementing countermeasures such as resource multiplication, which mitigates the effects of DDoS attacks [243], is necessary to ensure service continuity. Moreover, malicious adversaries can attack the localization infrastructure, e.g., by attenuating or amplifying signals [244], causing incorrect outputs. In the context of multi-floor localization, such attacks could mislead users to entirely incorrect floors [245], potentially compromising user safety. Detecting and mitigating these attacks, e.g., using anomaly detection techniques, is crucial [244].

2) PRIVACY

Location data is highly sensitive and must be protected against unauthorized access, misuse, and interference [2]. Users should maintain complete control over their location information, sharing it only with explicit consent. Technologies used for multi-floor localization vary in their level of privacy vulnerability. Specifically, technologies that require active sensing and constant data exchange with external infrastructure or server, such as common approaches based on WiFi, BLE, or cellular infrastructure-dependent fingerprinting and approaches relying on camera data, can inadvertently expose sensitive movement patterns unless properly secured, posing significant privacy vulnerabilities. On the other hand, passive sensing approaches that do not require active signal transmission, such as those using inertial sensors and ambient signal monitoring (e.g., barometers or magnetometers), provide greater inherent privacy preservation by design.

Recent advances in privacy-preserving frameworks for 2D localization offer promising techniques, including differential privacy, k-anonymity, ℓ-diversity, t-closeness, (α, k)-anonymity, and δ-presence [246], [247]. Adapting and extending these privacy-preserving frameworks to multi-floor localization systems, especially if data is stored or processed on untrusted servers without anonymization, is a promising research direction. In addition, designing edge processing mechanisms that limit raw data transmission and integrating user-controlled consent models are crucial for enhancing privacy in real-world deployments.

E. EMERGING TECHNOLOGIES-RELATED DIRECTIONS

1) RIS FOR DEVICE-FREE MULTI-FLOOR LOCALIZATION

Existing multi-floor localization systems require users to actively use a device, making them unsuitable for applications such as intrusion detection and tracking. Although specialized equipment such as surveillance cameras [184] can address this need, their deployment and maintenance incur additional costs. Device-free passive localization systems, demonstrated in 2D scenarios, leverage

TABLE 5. Performance comparison of the different multi-floor localization systems. Metrics are either reported in the reviewed papers or estimated based on technology characteristics and employed methods.

Technology	Challenge or Taxonomy	Method	Testbed	Accuracy	Latency	Computational Complexity	Deployment Cost
WiFi	Signal Space Ambiguity	Clustering based on RSS and spatial similarity [56]	Four-floor building with 4 m floor height and 108 m x 203 m area. Building includes two large hollow spaces with sizes 1150 and 1250 m^2	Exact floor estimation accuracy: 97.8%	Low (lightweight clustering and matching)	Low (lightweight clustering and matching)	High (requires full fingerprinting data collection)
		Feature mapping based on SAE with stacking model fusion [77]	Three buildings, each having 4 to 5 floors with a total area of 108,703 m^2 [262]	Exact floor estimation accuracy: 95.1%	Medium (relies on stacking DL models)	Medium (relies on stacking DL models)	High (requires full fingerprinting data collection)
		Normalization based on AP installation floor and RSS [49]	A three-floor shopping mall and two four-floor campus buildings including a large atrium and non-uniform AP density [262]	Exact floor estimation accuracy: 91.8%	Low (lightweight normalization and model)	Low (lightweight normalization and model)	Low (Calibration-free system with no fingerprinting data collection)
	Low Accuracy	RSS images processed using CNN after ELM-AE obtains salient features [102]	Ujiindoorloc [262] and Tampere [263] datasets, including one five-floor building and two four-floor buildings	Exact floor estimation accuracy: 96.3%; avg. error: 8.34 m	Medium (relies on CNN and ELM-AE)	Medium (relies on CNN and ELM-AE)	High (requires full fingerprinting data collection)
		RSS time series processed using a CNN [57]	A three-floor building and two four-floor buildings [262]	Exact floor estimation accuracy: 100%; avg. error: 2.77 m	High (relies on obtaining multiple consecutive RSS measurements)	Medium (time series processing and CNN)	High (requires full fingerprinting data collection)
		Deep reinforcement learning to hierarchically bisect the 3D search space [59]	16-floor building, including three basement levels, with a total area of 44,000 m^2 [75], and two four-floor buildings with a total area of 108,703 m^2 [262]	Median error: 0.39 m	Medium (relies on deep reinforcement learning)	High (relies on a large model)	High (requires full fingerprinting data collection)
	Calibration Data Collection Overhead	Crowdsourcing and clustering WiFi and barometer measurements [50]	A six-floor teaching building and a nine-floor library	Exact floor estimation accuracy: 96%	Low (lightweight clustering-based algorithm)	Medium (requires continuous data upload and processing)	Low (relies on crowdsourcing data collection)
		Semi-supervised learning using attention-based graph neural network to cluster signal readings [104]	152 buildings with floors ranging from three to ten, and two five-floor and one seven-floor shopping malls	Exact floor estimation accuracy: 88%	Low (lightweight clustering algorithm)	Medium (requires partial retraining)	Low (requires only one labeled signal sample)
		Data augmentation based on multi-output Gaussian process [101]	Three buildings, each having 4 to 5 floors with a total area of 108,703 m^2 [262]	Exact floor estimation accuracy: 93.8%; avg. error: 8.6 m	Medium (reliance on LSTM and AE for inference)	Medium (reliance on LSTM and AE for inference)	Medium (requires few collected samples to generate more samples by augmentation)
		Proximity-based heuristics for floor identification [52]	Two four-floor buildings and one three-floor building with areas 152,099, 34,213, and 22,197 m^2	Exact floor estimation accuracy: 95.3%; avg. error: 6.5 m	Low (heuristic-based)	Low (lightweight algorithm)	Low (no fingerprint data collection needed)
	Low Portability	Virtualization based on strongest AP [51]	A five-floor building with area 22,570 m^2 [263], and two four-floor buildings with areas 2725 and 7114 m^2 [262]	Exact floor estimation accuracy: 92%	Medium (relies on CNN)	Medium (relies on CNN)	Low (small retraining and redeployment overhead)
	Resource Requirements	Lightweight ELM-based approach [64]	Fifth to ninth floor of a research center with data collected in 1095 reference points	Exact floor estimation accuracy: 99.3%; avg. error: 3.63 m	Low (lightweight algorithm)	Low (lightweight algorithm)	High (requires full fingerprinting data collection)
		Model compression with CAE and CNN [55]	Three buildings, each having 4 to 5 floors with a total area of 108,703 m^2 [262]	Exact floor estimation accuracy: 93.9%, avg. error: 7 m	Low (relies on model compression designed specifically for resource-constrained devices)	Low (relies on model compression designed specifically for resource-constrained devices)	High (requires full fingerprinting data collection)
		Fingerprint shrinking through PCA and employ an ensemble of ELM models [87]	Seven-floor building	Exact floor estimation accuracy: more than 95%	Low (lightweight ELM)	Low (memory-efficient representation)	High (requires full fingerprinting data collection)

widely available infrastructures such as WiFi APs to detect and track users by monitoring signal attenuation caused by their bodies [248], [249]. Extending these concepts to multi-floor environments, where transmitters and monitoring points can be located on different floors, is an interesting research direction. Recent advances in reconfigurable intelligent surface (RIS) technologies provide a potential solution. RIS can enable monitoring points to receive signals from transmitters on other floors [250], allowing deployment of device-free multi-floor localization systems.

TABLE 5. *(Continued.)* **Performance comparison of the different multi-floor localization systems. Metrics are either reported in the reviewed papers or estimated based on technology characteristics and employed methods.**

Technology	Challenge or Taxonomy	Method	Testbed	Accuracy	Latency	Computational Complexity	Deployment Cost
Bluetooth	Signal Space Ambiguity	Fingerprinting-based adaptive weighted fusion algorithm [114]	Three-floor building with an atrium space structure	Exact floor estimation accuracy: 92.7%	Medium (relies on a short sequence of samples)	Low (lightweight algorithm)	High (requires BLE beacon deployment and fingerprint data collection)
		Proximity-based approach with maximum voting of highest BLE RSS, and employ AE to obtain robust latent space [6]	35 malls in 7 cities with an average of 5 floors and 49,000 m^2 per building, including atrium areas	Exact floor estimation accuracy: 96%; median error: 2.4 m	Medium (relies on AE and fully connected network)	Medium (relies on AE and fully connected network)	Medium (requires moderate beacon deployment density)
Cellular	Signal Space Ambiguity	Feature selection and KNN-based fingerprint matching [116]	Three buildings in different cities with nine, twelve, and sixteen floors	Exact floor estimation accuracy: 73%	Medium (relies on a short window of samples)	Low (KNN-based)	High (requires full fingerprinting data collection)
		Feature mapping using deep LDA to get a floor-distinguishing feature space, and then employs an LSTM [122]	Ten-floor shopping mall and five-floor campus building in different cities	Exact floor estimation accuracy: 93.8%	High (relies on a sequence of samples and deep models)	Medium (relies on deep models)	Medium (allows unseen floors during fingerprint data collection)
		Sub-space-based estimation for fingerprint matching using a dedicated model for each subspace [125]	Five high-rise buildings with 10, 11, 20, 24, and 24 floors	Exact floor estimation accuracy: 89.8%	Low (lightweight algorithm)	Medium (requires a model per subspace)	Low (relies on crowdsourcing data collection)
	Calibration Data Collection Overhead	Data augmentation using Gaussian noise injection and deep generative models along with a CNN for classification using only the serving tower signals [127]	Nine-floor shopping mall with area of 725 m^2 and seven-floor campus building with area of 3950 m^2	Exact floor estimation accuracy: 93.7%	High (relies on a sequence of samples and deep models)	Low (relies on a lightweight classifier design)	Medium (requires few collected samples to generate more samples by augmentation)
		Proximity-based and lateration-based floor identification using indoor femtocells [120]	Four femtocell base stations in a two-floor building	Exact floor estimation accuracy: 69.9%	Low (lightweight algorithm)	Low (lightweight algorithm)	High (requires dense indoor femtocell deployment for accurate performance)
	Low Accuracy	NLP-inspired method to represent cellular RSS and process it using a transformer-based model [128]	Nine-floor shopping mall with area of 725 m^2 and seven-floor campus building with area of 3950 m^2	Exact floor estimation accuracy: 99.9%	High (relies on a sequence of samples and large language models)	High (relies on large language models)	High (requires edge devices to run the inference)
Barometer	Barometer Environmental Sensitivity	Reference station air pressure readings integrated using an adaptive Kalman filter [133]	An 8-floor building and a 15-floor building with participants moving up and down the stairs	Avg. vertical positioning error: 0.86 m	Medium (Requires updates from reference station)	Medium (Uses continuous pressure updates and Kalman filtering)	High (Relies on a nearby reference station)
		Single reference barometer air pressure readings on a floor with known altitude are compared with phone barometer readings [136]	A 5-floor building and a 6-floor building with the reference barometer placed on the 3rd and 2nd floors respectively	Vertical positioning error: around 15 cm	Low (lightweight algorithm)	Low (lightweight algorithm)	Medium (requires an external reference barometer and measuring the pressure difference for each floor from the reference barometer)
		Crowdsourced calibration via a transitive technique by utilizing user encounters to share and propagate air pressure data [130]	Simulation and a field test in a 10-floor building	Exact floor estimation accuracy: 98%	High (requires continuous calibration to obtain accurate results)	Medium (requires infrastructure to propagate and process calibration data)	Low (relies on crowdsourcing data collection)
	Sensor Drift and Noise	Reset errors using nearby reference station readings based on the changes in the readings [131]	6-floor building.	Avg. vertical positioning error: 0.6 m	Medium (Requires updates from reference station)	Low (lightweight algorithm)	High (Relies on a nearby reference station)
Inertial Sensors	Special Hardware	Body-worn sensing module used for PDR, detecting walking behaviors, and stair counting [143]	Eight subjects walking around in a 2-floor building	Walking behavior detection accuracy (including floor transition detection): 96.8%	Low (lightweight algorithm)	Low (lightweight algorithm)	High (requires special body-worn sensing module per user)
	No Special Hardware (Smartphone-Based)	Smartphone accelerometer is leveraged based on user encounters to track floor traversals [140]	Simulation and a field study in a 10-floor building with two elevators	Exact floor estimation accuracy: 97%	High (requires convergence based on user encounters, which took around three hours in the field study)	Low (lightweight algorithm)	Low (no special hardware required)

2) LEO SATELLITES FOR GLOBAL MULTI-FLOOR LOCALIZATION

Low Earth orbit (LEO) satellite constellations, deployed by private companies such as SpaceX Starlink, operate at much lower altitudes than traditional global navigation satellite systems (GNSS) such as GPS, allowing their signals to penetrate indoor environments [251], [252]. Although used primarily for global communication and

TABLE 5. *(Continued.)* **Performance comparison of the different multi-floor localization systems. Metrics are either reported in the reviewed papers or estimated based on technology characteristics and employed methods.**

Technology	Challenge or Taxonomy	Method	Testbed	Accuracy	Latency	Computational Complexity	Deployment Cost
Hybrid	Barometer and Inertial Sensors	Improved landmark detection leveraging pressure readings for detecting floor changes fused with inertial sensor readings for multi-floor location tracking [148]	Four-floor building with area of 3300 m^2, with testing conducted over three floors only	Landmark detection accuracy: 97.5%; tracking error: less than 2 m 80% of the time	Low (lightweight algorithm)	Low (lightweight algorithm)	Low (no data collection overhead and uses existing smartphone barometer and inertial sensors)
		Improved activity recognition using accelerometer data to detect mobility states and adapt barometer readings [150]	4-floor, 9-floor, and 11-floor office buildings, and a 5-floor shopping mall	Exact floor estimation accuracy: 99%	Low (lightweight algorithm)	Low (lightweight algorithm)	Low (no data collection overhead and uses existing smartphone barometer and inertial sensors)
	WiFi and Barometer	Infrastructure-free floor change detection combining WiFi and barometer readings through a hidden Markov model and XGBoost [154]	Four buildings with 4, 3, 12, and 4 floors, having atriums and hollow spaces	Exact floor estimation accuracy: 99.2%	Low (continuous tracking allowing a latency of less than 2 seconds 90% of the time)	Medium (relies on Markov model and XGBoost fusion)	High (requires full WiFi fingerprinting data collection)
		Automatic barometer calibration by fusing WiFi fingerprinting with crowdsourced barometer readings using a Markov localization framework [152]	4-floor campus building, a 4-floor shopping mall, and an 8-floor residential building of 448 m^2, 11614 m^2, and 760 m^2 area	Exact floor estimation accuracy: 87.3%	Medium (simultaneous barometer calibration)	Medium (relies on Markov framework for periodic barometer recalibration)	Low (relies on crowdsourcing data collection)
	WiFi and Inertial Sensors	Infrastructure-free floor change detection leveraging inertial sensor readings for floor traversal activity recognition combined with WiFi fingerprints processed using LDA [163]	6-floor building	Exact floor estimation accuracy: 94.3%; avg. error: 1.2 m	Low (lightweight algorithm)	Low (event-based inference only during floor transition detection)	Medium (requires some WiFi fingerprinting data collection)
		Automatic inertial sensor calibration using crowdsourced WiFi fingerprinting data and a particle filter, allowing activity recognition and PDR for multi-floor localization [166]	Two 4-floor buildings	Exact floor estimation accuracy: 100%; median error: 3 m	Low (continuous tracking)	Mediun (reliance on a particle filter)	Low (relies on crowdsourcing data collection)
	WiFi, Barometer, and Inertial Sensors	Integrating inertial sensor PDR with WKNN-based WiFi fingerprint matching through an adaptive particle filter [170]	A 3-floor building and a 2-floor building	Tracking error: less than 1.98 m 75% of the time	Low (continuous tracking)	Medium (relies on adaptive particle filter)	Medium (requires some WiFi fingerprinting data collection)
		Process time series of WiFi and inertial sensor measurements using LSTM, integrating barometer readings through Bayesian classification for floor identification [173]	3-floor building	Avg. error: 0.42 m	Medium (initially relies on obtaining consecutive measurements)	Medium (relies on LSTM)	Medium (requires some WiFi fingerprinting data collection)
		PDR trajectory optimization framework that combines WiFi finite-time measurement anchors with inertial and barometer sensors using unscented Kalman filter [178]	2 floors from a building with 3000 m^2 area	Tracking error: less than 1.38 m 75% of the time	Low (continuous tracking)	Medium (relies on unscented Kalman filter)	Low (no fingerprinting data collection required)
Other Technologies	Technology-Specific Limitations	Requires additional infrastructure costs to fuse BLE signals with inertial and barometer sensors using adaptive unscented Kalman filter [183]	2 floors from a building	2D positioning error: 1.08 m; vertical positioning error: 0.46 m 75% of the time	Low (continuous tracking)	Medium (relies on unscented Kalman filter)	High (requires dense BLE beacon deployment)
		Combining inertial sensors with smartphone camera using CNN networks and PDR [184]	Three floors of a building with a walking trajectory of 42.64 m	Tracking error: less than 0.9 m 95% of the time	Medium (relies on CNN)	High (requires smartphone camera)	Medium (requires obtaining reference photos at each floor)
		Integrating magnetic fingerprints and inertial sensor readings with KNN for fingerprint matching [185]	Simulation in 10-floor, 8-floor, and 6-floor buildings	Exact floor estimation accuracy: 89.3%	Low (lightweight algorithm)	Low (lightweight algorithm)	High (requires full magnetic fingerprinting data collection)

Internet services, LEO satellite signals have shown potential for opportunistic outdoor positioning in GNSS-deprived regions [253]. An interesting research direction is to study the feasibility of LEO-based multi-floor localization, leveraging their ability to penetrate indoor areas. This approach could pave the way for an infrastructure-free and globally ubiquitous multi-floor localization system that works in any building worldwide, similar to how GNSS provides

global outdoor localization [251]. A potential approach is to build a global database that maps buildings, floors, and locations worldwide to specific LEO satellite signal fingerprints [254].

3) QUANTUM MULTI-FLOOR LOCALIZATION

Quantum computing, with its ability to leverage superposition and entanglement phenomena [255], offers computational efficiency far beyond classical methods, especially for complex problems. As quantum computing matures and corporations work toward real-world applicability [256], its potential for designing efficient localization algorithms grows [257]. Recently, a quantum fingerprinting-based 2D localization algorithm demonstrated exponential improvements in space and runtime over classical approaches [258]. Extending quantum algorithms to large-scale multi-floor environments with several transmitters could significantly enhance efficiency and scalability of multi-floor localization.

VII. CONCLUSION

In this paper, we provided a comprehensive survey of multi-floor localization systems, emphasizing domain-specific challenges, solutions, and recent advances. We presented the central challenges in implementing and deploying multi-floor localization systems, categorizing them into input-level and system-level challenges. We then provided an extensive review of various multi-floor localization systems, organized by the technologies employed, alongside an analysis of their proposed solutions for addressing these challenges and a comparison of their performance trade-offs. Moreover, we highlighted open research problems and future directions, including approaches to overcoming challenges posed by complex environmental structures, reducing deployment overhead, addressing user-related and device-related variations, improving security and privacy, and leveraging emerging technologies. We complemented these discussions by exploring applications that underscore the importance of multi-floor localization and by providing background on how existing localization techniques are extended to multi-floor scenarios.

Although significant progress has been achieved in indoor localization, multi-floor localization remains a fertile area for research and innovation. Further experimentation and integration of novel technologies is necessary to develop more accurate, scalable, efficient, and practically deployable multi-floor localization systems.

APPENDIX A APPENDIX
EXTENDED SYSTEM PERFORMANCE COMPARISON TABLE

Table 5 shows a comprehensive performance analysis corresponding to Table 4 and Figure 13 discussed in Section V-H.

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